Supply Chain Analytics 15 (2026) 100239

Contents lists available at ScienceDirect

Supply Chain Analytics

journal homepage: www.sciencedirect.com/journal/supply-chain-analytics

A human-centred analytics framework for auditable supply chain risk monitoring and decision support

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ARTICLE INFO

Keywords:
Large language models
Risk analytics
Rule-violation detection
Human-centred decision support
Edge computing
Multi-agent analytics

ABSTRACT

Supply chain managers must monitor multi-domain logistics data continuously, detect emergent risks and act faster than manual analysis allows. Large Language Models (LLMs) reason contextually but carry latency and hallucination risk unsuitable for continuous detection. Three gaps persist: AI-enabled supply chain risk monitoring remains predominantly retrospective; no framework implements hybrid LLM-SLM (Small Language Model) deployment in multi-agent architectures or combines rule-based detection with practitioner-facing natural language decision support. This paper addresses these gaps through a human-centred, hierarchical three-tier multi-agent framework that separates detection from explanation: deterministic rules handle rule-violation detection and risk scoring; SLM worker agents, invoked only when rule violations are confirmed, generate domain-specific summaries; and a supervisor LLM synthesises cross-domain findings into a structured risk report. The human operator retains full decision authority at the third tier. Because detection is deterministic and rule-based, it is fast and verifiable, and hallucination is excluded from the detection layer by construction; residual hallucination risk is confined to the generative explanation layer, where human review remains necessary before any operational action. Evaluated on a 32,065-record logistics dataset, the framework screens eight SLM architectures (1–7 billion parameters) and benchmarks four finalists on 100 held-out records. The preferred candidate, Granite 3.1 MoE 3B, reaches 112.2 tokens per second on edge hardware (6 GB VRAM). Compression loss, the omission of items carried by the reference report, accounts for 86.6 per cent of classified divergences. At scale (1000 records), hybrid edge-cloud configuration reduces processing time by 57 per cent and API cost by 39.6 per cent versus fully cloud-based deployment. The evaluation is batch-mode proof of concept; live-stream validation and practitioner user study remain future work. The paper contributes to data analytics through rule-based detection, composite risk scoring, and multi-model benchmarking; and a hybrid edge-cloud AI architecture; and to decision support across supply chain.

1. Introduction

Supply chain managers face decision-making demands that have grown sharply in both urgency and complexity. Modern supply chains, whilst economically efficient through their global integration, create propagation pathways along which localised disruptions cascade across networks of suppliers, manufacturers, and logistics service providers, producing systemic ripple effects [1–3]. Industry 4.0 logistics intensifies this further: cyber-physical infrastructure, IoT-enabled data streams, and real-time monitoring have raised the expectation of continuous responsiveness from a competitive advantage to an operational necessity [4–6]. Where traditional analytical workflows require days or weeks to move from data to decision, LLM-enabled approaches can compress that cycle to minutes or hours [7], strengthening the operational case for automated decision support.

Even where automation is deployed to bridge this gap, the constraint shifts rather than dissolving. Current logistics networks already produce thousands of hourly observations from vehicle global positioning systems (GPS) trackers, IoT-enabled cargo sensors, warehouse management systems, and external transport providers [8,9]; the difficulty lies in interpreting these streams transparently and quickly enough that practitioners can act on the resulting insights with justified confidence.

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https://doi.org/10.1016/j.sca.2026.100239
Received 31 May 2026; Received in revised form 23 July 2026; Accepted 24 August 2026
Available online 27 August 2026
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Drawing on a survey of 1770 managers across fourteen countries, Kolbjørnsrud et al. [10] report that whilst 84 per cent of respondents expected AI to make their work more effective and interesting, only 14 per cent of front-line managers strongly agreed that they would trust AI advice in making business decisions, precisely those managers most directly responsible for operational risk mitigation. The same study identifies three conditions for trusting AI-generated advice: understanding how the system works and generates its outputs (cited by 61 per cent of respondents), evidence of a proven track record (57 per cent), and the system's ability to provide convincing explanations (51 per cent). The trust deficit underlying these figures is rooted in the opacity of many AI systems: practitioners cannot scrutinise, contest, or understand the rationale underlying recommendations that cannot be traced back to explicit decision logic, and systems that cannot be interrogated in this manner tend to remain untrusted and underutilised [11,12].

Existing approaches to AI-enabled decision support for operational logistics monitoring each present a distinct trade-off between detection reliability [9], contextual reasoning [13], and the conditions for practitioner trust [11,14]. Traditional rule-based systems offer transparency through their explicit, hand-engineered logic but struggle to capture the contextual and cross-domain dependencies needed for multi-domain anomaly assessment [15,16]. Machine learning and deep learning models have shown strong predictive accuracy across demand forecasting, supplier risk modelling, and operational disruption prediction [17], but when used as the primary decision mechanism, their opacity reinforces the trust deficit that impedes adoption [11,12]. Large Language Models (LLMs) have demonstrated capability in strategic, retrospective supply chain tasks such as automated risk identification from news and supplier databases, scenario-based disruption assessment, and natural language interaction with optimisation tools [7,18,19]. That capability comes with an important constraint. LLMs are probabilistic and carry measurable inference latency; they are well suited to deliberative, offline reasoning tasks but poorly suited to the real-time latency requirements of continuous operational monitoring [20,21]. Detecting that a shipment has deviated from its planned route, that a supplier has missed consecutive delivery windows, or that cargo temperature has exceeded its specified range requires fast, verifiable classification rather than inference from a language model. For such tasks, LLM latency imposes an operational cost that probabilistic outputs do not offset, and hallucination risk introduces the possibility that plausible but incorrect outputs propagate into time-critical operational decisions, a concern that Boone et al. [22] identifies as warranting hybrid human-AI oversight mechanisms in supply chain contexts.

Resolving this tension rests on an architectural separation between the detection task and the explanation task. Three developments make this separation possible in practice. First, agentic artificial intelligence has matured as a framework for hierarchical, multi-domain problem solving, in which specialised agents execute narrowly scoped monitoring tasks whilst a supervisory agent performs cross-domain integration and synthesis [23–25]. Second, the heterogeneous deployment of large and small language models has proven both technically and economically viable, enabling models to be matched to tasks rather than relying uniformly on large cloud-based systems [20,21]. Small Language Models (SLMs) have been shown to match or surpass larger models on specific, narrowly scoped tasks whilst requiring only a fraction of the computational resources, with self-hosted SLMs delivering cost reductions of 5–29 times relative to cloud-based LLM APIs [21]. Third, research on transparency in human-AI collaboration shows that trust increases when a system's reasoning processes are made explicit through auditable logical structures or natural language justifications and counterintuitively decreases when systems foreground their probabilistic uncertainty [11]. Yan et al. [12] argue further that human-centric AI systems must be designed to augment rather than replace human capabilities, with transparency, adaptability, and ethical alignment as core principles, requirements that are structurally unmet by both opaque machine-learning classifiers and non-deterministic large language models. These principles are central to the Industry 5.0 paradigm, which reorients operations and supply chains around resilience, sustainability, and human-centricity rather than technology-driven automation alone [12,26]. Its extension to supply chain management has been conceptualised as Supply Chain 5.0, in which supply chains are redesigned as resilient, sustainable, and human-centric systems [27]; within this orientation, AI is positioned to augment rather than replace human judgement, preferring monitoring architectures in which detection remains auditable and decision authority rests with the human operator.

Three gaps remain in the existing research. First, AI-enabled SCRM is treated predominantly as a deliberative, retrospective task; no framework currently demonstrates detection and risk synthesis at the latency that continuous operational monitoring requires. Second, no framework puts task-differentiated hybrid LLM-SLM deployment into practice within a hierarchical multi-agent architecture. Third, no framework combines rule-based detection with practitioner-facing natural language decision support, leaving open two of the five priority directions for AI-based SCRM identified by Ordibazar et al. [14]: explainability methods and practitioner facing dashboards.

This paper addresses these three gaps through a hierarchical three-tier multi-agent framework that separates deterministic rule-violation detection from language-model-based explanation, evaluated empirically on a logistics dataset spanning road, drone, and rail transport modes. Four contributions follow. First, the framework formalises an architecture in which deterministic rules perform rule-violation detection and language models are confined to a secondary explanation layer, an arrangement designed for continuous operational monitoring and demonstrated here in batch mode as a proof of concept; this separation, to the best of our knowledge, has not been made explicitly in prior frameworks [23]. Second, it provides empirical evidence that selecting models to match the demands of each agent's task, rather than defaulting to the largest available model, produces a measurable performance advantage at the system level, extending the position of Belcak et al. [20] with evidence from a supply chain risk monitoring context. Third, it shows that auditable, rule-based detection paired with natural language explanation can address the three trust conditions identified by Kolbjørnsrud et al. [10] and supported by Vössing et al. [11], offering an alternative to the post-hoc explainability methods most common in the current literature [14] and responding to the human-centric AI principles articulated by Yan et al. [12]. Fourth, the empirical evaluation yields a cost and latency comparison between hybrid edge-cloud and fully cloud-based deployments, offering a reference point for those evaluating local-SLM worker tiers in continuously operating monitoring systems.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature across AI-enabled decision support in supply chain management; real-time monitoring and anomaly detection; agentic AI and multiagent architectures for supply chains; and transparency, trust, and human-centred AI in operational decision support, establishing the three research gaps that motivate this study. Section 3 describes the dataset, framework architecture, and evaluation protocol. Section 4 reports the experimental results. Section 5 discusses the findings, theoretical contributions, practical implications, and limitations. Section 6 presents the conclusion and directions for future research.

2. Relevant literature

2.1. AI-enabled decision support in supply chain management

The integration of AI into supply chain management is accelerating across functions including demand forecasting, inventory management, supplier selection, and risk monitoring [5,17,28]. The economic contribution of AI technologies to the global economy is projected to reach approximately USD 13 trillion by 2030 [29]. Generative AI in particular has seen a sharp upward trend in supply chain applications,

with fewer than five papers published per year before 2022 and 45 published in 2024 alone [30]. Recent bibliometric analysis identifies explainable AI, human-AI collaboration, and scalable deployment among the key future research directions in this domain, alongside data management, robustness, and real-time optimisation [31], signalling that the field has moved beyond capability demonstration towards questions of practical adoption and trustworthy deployment.

Barriers to adoption persist at multiple levels. Cannas et al. [28], in a multiple case study across six organisations, identify barriers at technical, organisational, and financial levels, including data quality challenges, shortage of domain-specific skills, high investment requirements, and insufficient clarity on economic benefits as the primary obstacles to AI implementation in supply chain and operations management. Bahroun et al. [30], in their PRISMA-guided systematic review of 98 studies, find that most evidence for GenAI in supply chain management remains at the prototype level and rarely reports system-wide key performance indicators, indicating that the field has yet to bridge the gap between capability demonstration and operational deployment. Whilst Bahroun et al. [30] identify this maturity gap, Callefi et al. [32] show that data integration is the foundational enabling capability for GenAI-driven supply chain value, whilst adaptability and strategic differentiation emerge as dependent capabilities at later stages of capability development.

At the sociotechnical level, a further challenge concerns the relationship between AI systems and the practitioners who must act on their outputs. Hendriksen [33] argues that AI integration in supply chains is not only a technical but also a sociotechnical process shaped by human sensemaking, and that the trajectory of AI adoption depends critically on how practitioners interpret, make sense of, and develop trust in the systems they are asked to work with. The trust deficit and its specific conditions were detailed in Section 1 (drawing on [10] and [11]); what Hendriksen adds is the argument that these conditions are not static requirements to be met once but ongoing interpretive processes through which practitioners continuously negotiate their relationship with AI systems. Two architectural responses to this trust challenge have emerged in the literature. Yan et al. [12] argue that human-centric design principles, including transparency, adaptability, and ethical alignment, must be integral to system architecture rather than added as supplementary explainability layers. Boone et al. [22] call for human-in-the-loop frameworks and governance mechanisms to support responsible GenAI deployment, a position reinforced by Bahroun et al. [30], who identify governance-by-design as a priority for scalable adoption.

2.2. Real-time monitoring and anomaly detection in supply chain operations

The technical infrastructure for continuous operational monitoring has matured considerably, driven by the proliferation of IoT devices, GPS tracking systems, and sensor networks that generate continuous data streams across logistics operations. Several systems now demonstrate that anomaly detection at operational scale is feasible. GPS, radio frequency identification (RFID), and computer vision integration enables automated anomaly detection and transportation status monitoring in logistics operations [8], time-series methods provide early warning capabilities for abnormal demand fluctuations and delivery irregularities in large-scale supply chain transaction data [9], and digital twin frameworks achieve end-to-end visibility through the integration of real-time sensor, RFID, and enterprise resource planning (ERP) data with simulation and optimisation models, enabling proactive disruption risk management [4]. More recent work extends this infrastructure: Chan and Siqin [34] examine the integration of generative AI with Internet of Behaviors data for supply chain monitoring, showing how real-time sensor data can improve prediction, optimisation, and personalisation across supply chain operations. At the same time, IIoT integration introduces new vulnerabilities; Saran and Kesswani [35] find that whilst IIoT substantially enhances the functionality, visibility, and decision-making capacity of supply chain management systems, it also amplifies cybersecurity exposure through the interconnection of operational components across the supply chain network.

These systems establish that detection at scale is technically achievable. A common feature across these frameworks, however, is that detection outputs are structured, quantitative, and machine-readable rather than human interpretable. The systems identify that an anomaly has occurred and classify it, but do not generate natural language explanations that practitioners can read, interrogate, and act on directly. This gap between detection capability and practitioner-facing decision support is ultimately a trust problem. Ordibazar et al. [14], in a systematic review of 72 AI-based SCRM papers organised across the four SCRM stages of identification, assessment, mitigation, and monitoring, identify monitoring as the least developed stage in the current literature (9 of 72 papers) and name five priority directions for future research: consideration of the ripple effect of risks, development of methods to explain AI-based SCRM models, capture of external event impacts, unified treatment across all SCRM stages, and the design of user-friendly practitioner dashboards. The absence of explainability across all SCRM stages is noted as a recurring theme, with risk managers requiring the ability to understand how AI-generated risk analyses are produced before they can act on them with confidence. The gap between automated detection and human-interpretable decision support at the monitoring stage is the operational problem that motivates the framework proposed in this study: anomaly detection without natural language explanation produces outputs that practitioners cannot interrogate, reinforcing the trust deficit identified in Section 2.1.

2.3. Agentic AI and multi-agent architectures for supply chain decision support

Multi-agent system frameworks have been applied to supply chain coordination since the early 2000s, using the principle of decentralised decision-making in which multiple autonomous agents interact to produce coordinated system-level outputs [25]. Xu et al. [25] present an autonomous supply chain system using a multi-agent approach that automates the full process from sourcing to delivery, distinguishing their work by adopting a software engineering methodology and implementing an actual software prototype rather than a simulation. Whilst this establishes the technical feasibility of MAS for supply chain automation, it operates without language model integration and relies on rule-based agent decision-making, limiting its ability to produce human-interpretable outputs. Yang et al. [36] apply multi-agent deep reinforcement learning for integrated demand forecasting and inventory optimisation in sensor-enabled retail supply chains, achieving an 18.2 per cent reduction in forecast error and a 23.5 per cent reduction in stockout rates compared with the baselines, but again without natural language explanation capabilities for operational managers. Aylak [16] present SustAI-SCM, an agentic AI framework for sustainable supply chain process automation using a transformer-based model, reporting a 28.4 per cent reduction in cost and a 30.3 per cent reduction in emissions, but focusing on automated execution rather than human-facing decision support.

Agentic AI research is now converging with advances in large language models, producing a new generation of LLM-powered agentic systems [23,24]. In a multi-expert analysis of AI agents and agentic systems, Hughes et al. [24] identify supply chain disruption prediction and response as one of the most promising application domains for agentic systems, whilst flagging hallucinations, transparency, bias, privacy and security as the primary barriers to adoption. Boskabadi et al. [23] propose an industrial agentic AI architecture, Sense-Infer-Control (SIC), that integrates generative models within agentic ensembles for real-time process monitoring and decision-making across manufacturing and supply chain processes. Their four industrial case studies establish that the combination of deterministic sensing with generative inference

is technically viable in complex operational settings.

At the intersection of LLMs and multi-agent architectures for SCRM, Quan et al. [37] propose MARS, a multi-agent risk assessment framework that integrates LLMs to interpret unstructured severe weather narratives alongside structured risk indicators for logistics hub site selection. Their work confirms that LLM-enhanced multi-agent architectures can combine structured and unstructured risk factors into interpretable rankings. Zhao et al. [19] build a framework that uses LLMs to automate risk identification from news feeds and supplier databases, addressing the constraints of limited human resources and time that characterise traditional SCRM intelligence gathering. Their integration of a structured taxonomy to organise LLM-generated risk data for end-user interpretation represents a step towards practitioner-facing decision support, though the framework operates at the strategic risk identification level rather than at continuous operational monitoring. A case study by Zheng et al. [38] integrates LLMs with a decentralised agent-based system for supply chain tasks, showing how LLMs function as a natural language conversational interface through which practitioners can interact with specialised prediction agents, whilst noting that LLM reliance introduces challenges of computational demand and training cost that may constrain real-time performance.

SLMs have emerged as a viable and economically attractive alternative to large cloud-based models for resource-constrained or latency-sensitive applications. The key principles behind SLMs are outlined by Zhang et al. [39], who note that their advantages in efficiency, latency, safety, and privacy make them particularly suitable for deployment on resource-constrained devices and in real-time applications where LLM latency is unacceptable. In a comprehensive survey of 70 open-source SLMs in the 100 M to 5B parameter range, Lu et al. [40] report that these models are already deployed at scale in commercial devices and perform well across a range of reasoning and language tasks, offering a technically mature foundation for resource-efficient agentic deployment. Cost reductions of 5–29 times are reported by Irugalbandara et al. [21] when cloud-based LLM APIs are replaced with self-hosted open-source SLMs in production environments, providing quantitative evidence for the economic viability of SLM deployment. Belcak et al. [20] argue that SLMs are sufficiently capable, inherently more suitable, and more economical for the specialised, repetitive tasks that characterise agentic worker agents, and that heterogeneous architectures combining SLMs and LLMs represent the natural choice when both specialised and general-purpose capabilities are required within the same system.

2.4. Research positioning

The proposed framework addresses the three gaps identified in Section 1. Table 1 maps each gap to the limitation of prior work and the corresponding design response of the three-tier architecture: deterministic rule-violation detection at Tier 1 on logistics sensor data, SLM and LLM generative synthesis at Tier 2 with models selected to match each tier's requirements, and a proposed human decision layer at Tier 3.

Table 1
Research gaps identified in the literature and the corresponding design responses of the proposed framework.

Gap	Problem statement	Limitation of prior work	This paper
Operational monitoring latency	No framework integrates sensor-level rule-violation detection with multi-domain generative synthesis within latency requirements consistent with continuous operational monitoring, producing natural language decision support.	LLM-based frameworks are strategic and offline [7,19,37]; continuous monitoring systems lack natural language explanation [8,9].	Hierarchical three-tier MAS with deterministic Tier 1 detection on logistics sensor data and conditional Tier 2 generative synthesis producing practitioner-facing natural language risk intelligence.
Hybrid edge-cloud deployment	Monolithic cloud-based LLM deployment is cost- and latency-prohibitive for continuously operating risk monitoring at scale.	SLMs reduce costs substantially relative to cloud-based APIs [21] and are argued to be more suitable for specialised agentic tasks [20], yet no framework puts hybrid LLM-SLM deployment into practice within a hierarchical multi-agent architecture.	Worker agents paired with locally deployed SLMs for high-frequency domain-specific summarisation; LLM reserved for supervisor-level cross-domain synthesis. Empirically evaluated across eight SLM architectures against a cloud-based baseline.
Transparency and decision-support	Practitioners will not act on AI-generated recommendations they cannot trace to explicit logic [10]; explainability of AI-based SCRM models and practitioner dashboards are flagged among five priority research directions [14].	ML classifiers produce opaque outputs; LLM reasoning is probabilistic and prone to hallucination, making it unauditable as a primary detection layer [22]. Hybrid human-AI oversight mechanisms remain underdeveloped [12,22].	All risk flags and scores derive from explicit, inspectable deterministic rules at Tier 1; language models only articulate confirmed findings. Complete traceable report delivered to human operator at Tier 3 (future step).

3. Research methodology

3.1. Dataset and data characterisation

A publicly available, anonymised dataset of 32,065 continuous hourly records, collected over a three-year period (January 2021 to January 2024) from a high-volume logistics network in Southern California [41], provides the empirical basis for this study. The records capture operational supply chain conditions across vehicle GPS trackers, IoT-enabled cargo sensors, internal warehouse management systems (WMS), and external data providers, spanning road haulage, drone delivery, and rail freight.

Of the 25 features in the dataset, four (disruption_likelihood_score, delay_probability, risk_classification, and delivery_time_deviation) are explicitly predictive target variables. Retaining them would introduce target leakage and reduce the framework to a supervised classifier rather than a proactive rule-violation detection system; they were therefore excluded from the input space. The remaining 21 features, verified to be structurally complete with no missing values, form the input matrix used throughout this study.

Visual inspection of distribution patterns across the 21 retained variables reveals pronounced skewness (Fig. 1). Fuel consumption rates, equipment availability, and traffic congestion level are concentrated at their nominal baselines, with rare deviations appearing as sparse long tails, a pattern consistent with the expected rarity of supply chain anomalies in which the majority of operational states remain within safe, non-disruptive thresholds.

Pearson correlation analysis across the 21 retained features (Fig. 2) finds no meaningful linear association between any pair. All off-diagonal coefficients fall below 0.015 in absolute value; the strongest is −0.015, between traffic congestion level and loading and unloading time, and only 13 of the 210 unique feature pairs exceed 0.01. With $n = 32{,}065$ records the standard error of a correlation under independence is

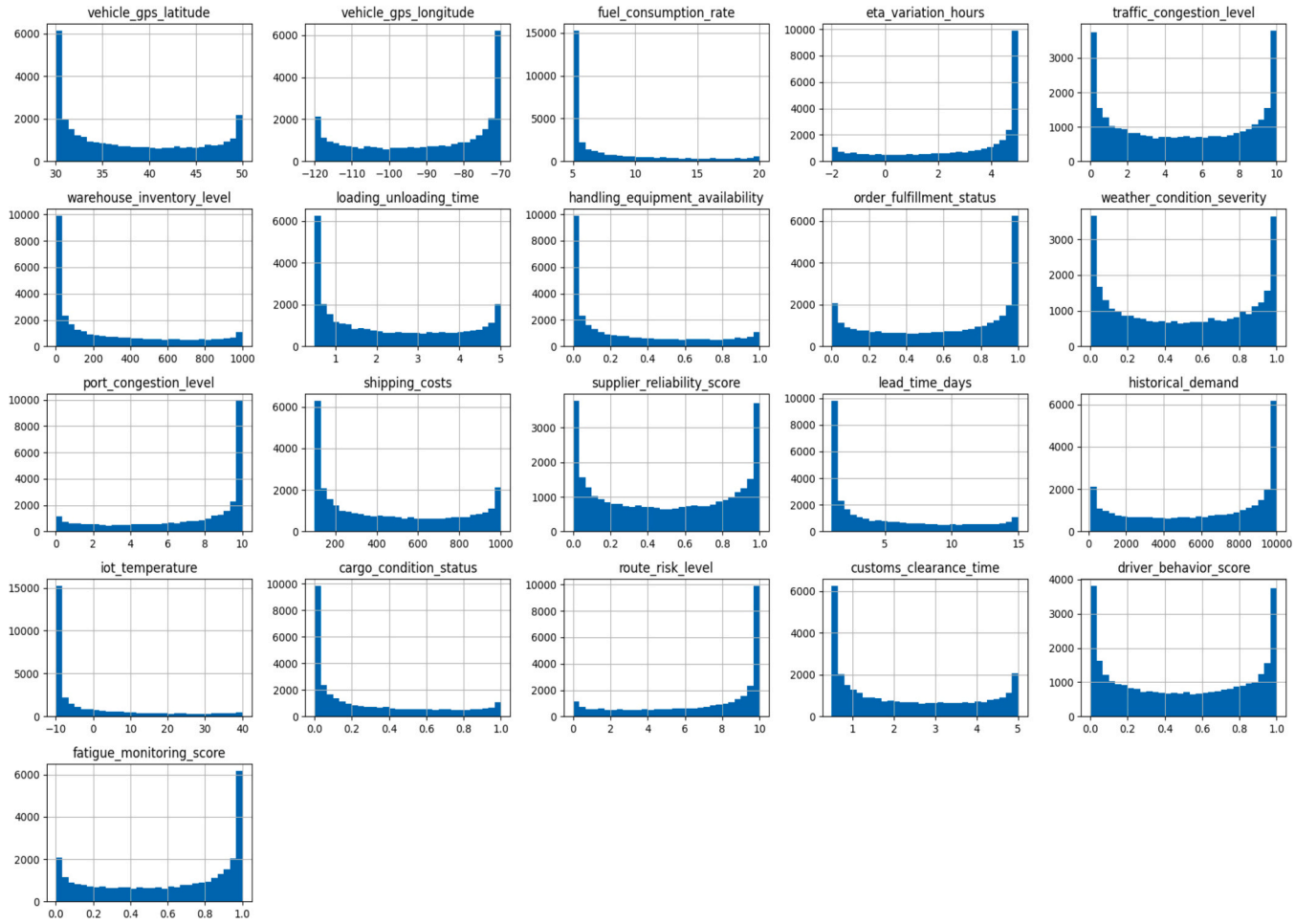


Fig. 1. Distribution of the 21 retained operational features, showing pronounced skewness across the majority with data concentrated at nominal baselines.

approximately 0.006, and the largest observed coefficient lies within the range that sampling variation alone would produce, so the features are statistically indistinguishable from mutually independent at the linear level. This is atypical for live operational logistics data, in which weak cross-domain dependencies, for example between traffic congestion, ETA variation, and route risk, are normally present. The dataset is documented by its publisher as anonymised and processed for privacy [41]; the absence of linear structure is consistent either with a decorrelating transformation applied during anonymisation or with independent feature generation, and the released data do not allow these to be distinguished. The framework makes no use of inter-feature correlation, so this property does not affect its operation, but it does limit the operational realism of the evaluation, which should be read as a proof of concept on a public, benchmark-style logistics dataset rather than a test under fully representative field conditions. Non-linear relationships are not captured by Pearson correlation and cannot be excluded on this evidence.

3.2. Architectural framework

The framework is a three-tier multi-agent pipeline in which each agent's logic, monitored features, contribution weights, and underlying language model can be configured independently, allowing the system's risk tolerance and computational profile to be adjusted without redesigning the pipeline. The empirical evaluation covers the two automated tiers only; the human decision layer is proposed as a design component for future validation.

3.2.1. Three-tier workflow architecture

Fig. 3 presents the three-tier pipeline as implemented in this study, with six specialised worker agents mapped to the operational domains and input features detailed in Table 2.

Tier 1: Deterministic Detection. Each processing cycle begins with a deterministic rule engine that performs rule-violation detection and risk scoring. In this framework, detection means checking incoming values against predefined operational thresholds, and a flagged event is a rule violation, not a statistically learned outlier, and the term rule-violation detection is used throughout to reflect this. Predefined business rules and quantitative thresholds are applied to incoming data, flagging rule-violations and computing a normalised risk score on a continuous scale of $s_i \in [0,1]$. No language model is involved at this stage; detection is fast, verifiable, and based entirely on domain-specific operational knowledge. Language models in subsequent tiers generate explanations and recommendations but do not independently detect or classify risk events.

Tier 2: Generative Synthesis. Two generative components operate at different scopes within this tier. At the worker level, the SLM paired with each agent is invoked conditionally, only when the deterministic tier has confirmed at least one rule violation in that domain. Where no threshold is breached, the agent produces its structured result directly from the deterministic output and the SLM is not called. When a rule violation is confirmed, the SLM receives the structured findings as a prompt and generates a concise, human-readable summary contextualised against operational norms, so that every generative output traces back to a confirmed deterministic finding. An SLM rather than a general-purpose LLM is chosen at this level because each agent's

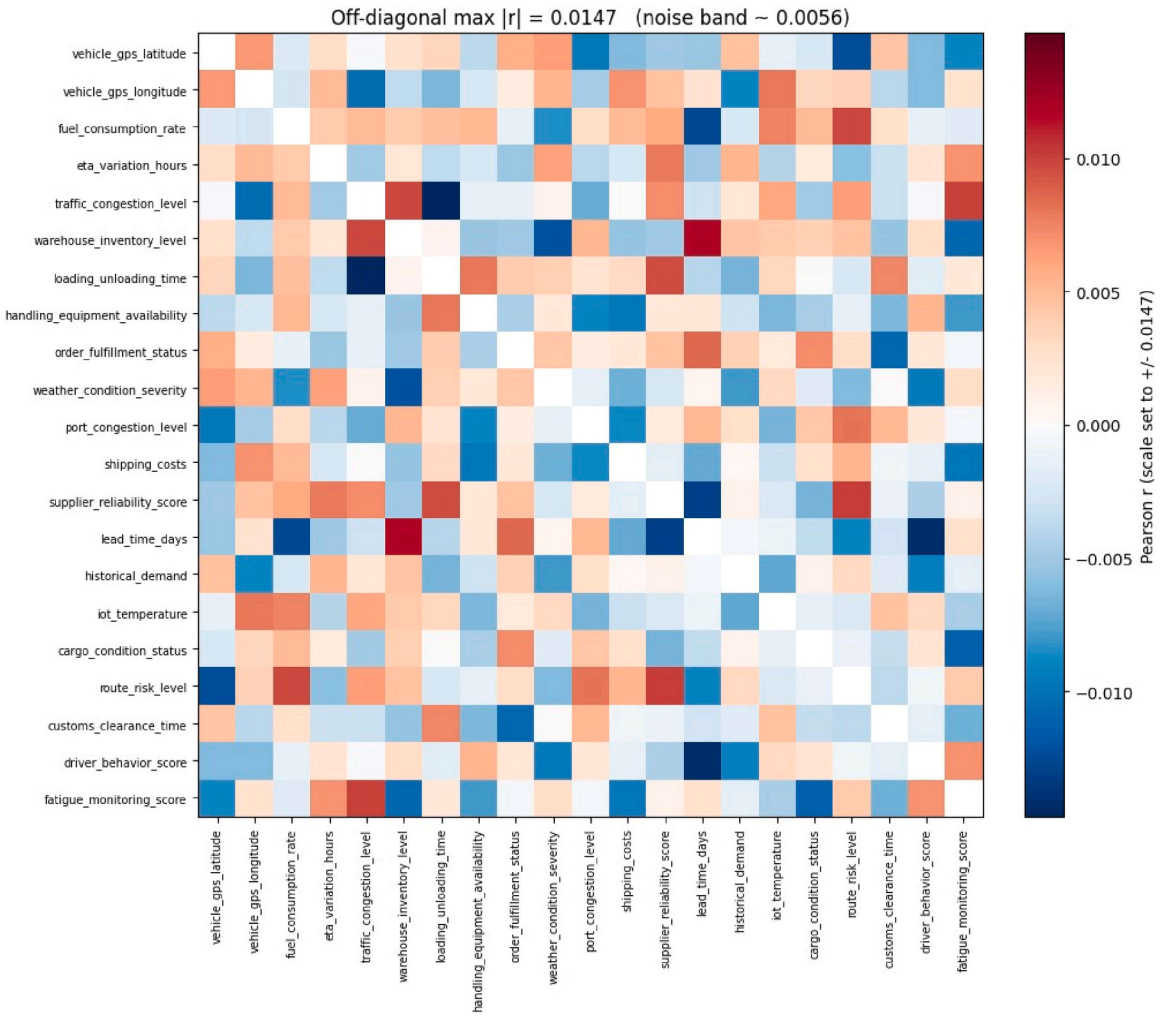


Fig. 2. Pearson correlation matrix of the 21 operational features. The colour scale is set to the off-diagonal range (± 0.0147) so that residual structure is visible, and the diagonal is omitted. All off-diagonal coefficients lie within sampling variation of zero (maximum |r| =0.015), indicating no meaningful linear association between feature pairs.

generative task is bounded to explaining a single anomaly within one domain. As Belcak et al. [20] argue, narrow, templated, domain-specific tasks are well served by SLMs, which offer lower inference latency, reduced computational cost, and tighter behavioural alignment than general-purpose LLMs. Zhang et al. [39] reinforce this position, noting that SLMs trained on domain-specific corpora complement rather than replace LLMs. Because the SLM is called only when an anomaly is confirmed, generative cost scales with the anomaly rate rather than with data volume.

At the supervisor level, a cloud-hosted LLM synthesises cross-domain findings from all six worker agents into a consolidated assessment and a set of prioritised recommendations. Unlike the worker-level task, the supervisor reasons across six operationally distinct inputs simultaneously, each with its own risk characteristics and terminology. This generalised, multi-source reasoning task falls outside the scope for which domain-specific SLMs are designed [20], making an LLM the appropriate choice. A global risk score is computed using the domain weights in Table 2 and mapped to a three-class classification: Low, Moderate, or High risk. Formal aggregation and classification thresholds are defined in Section 3.2.3.

Tier 3: Human Decision Layer. Under the proposed design, the complete risk report is delivered to a human operator who applies contextual judgement and retains full decision authority over any mitigation action. No automated action is executed at any stage; the system operates strictly as a decision support instrument. Validation of Tier 3 in a live operational setting is a primary objective for subsequent research. Its design draws on the finding by Vössing et al. [11] and Kolbjørnsrud et al. [10] that practitioners will not act on AI-generated recommendations they cannot trace to explicit logic, a concern addressed by making every risk flag and score auditable to its originating rule.

Fig. 4 maps these three tiers onto a physical deployment stack. Five layers, from data ingestion through to the worker SLM tier, execute on local edge-representative hardware; the supervisor LLM tier runs as a single cloud-hosted API call. Phase 3 of the experimental design (Section 3.3.1) benchmarks this hybrid edge-cloud configuration against a fully cloud-based baseline to quantify the practical advantage of local SLM deployment.

3.2.2. *Specialist worker agents*

Six worker agents are deployed, each assigned exclusive authority over a distinct functional domain. Effective supply chain decision-making requires a clear separation of decision authority across organisational functions, and this decomposition reflects that principle. As Xu et al. [25] observe, problem decomposition in multi-agent systems must respect organisational boundaries, distinguishing between structural agents that represent business entities and functional agents that manage specific operational tasks. Collapsing heterogeneous features into a single generalised agent would produce a system in which no output maps cleanly onto a decision that any one organisational function

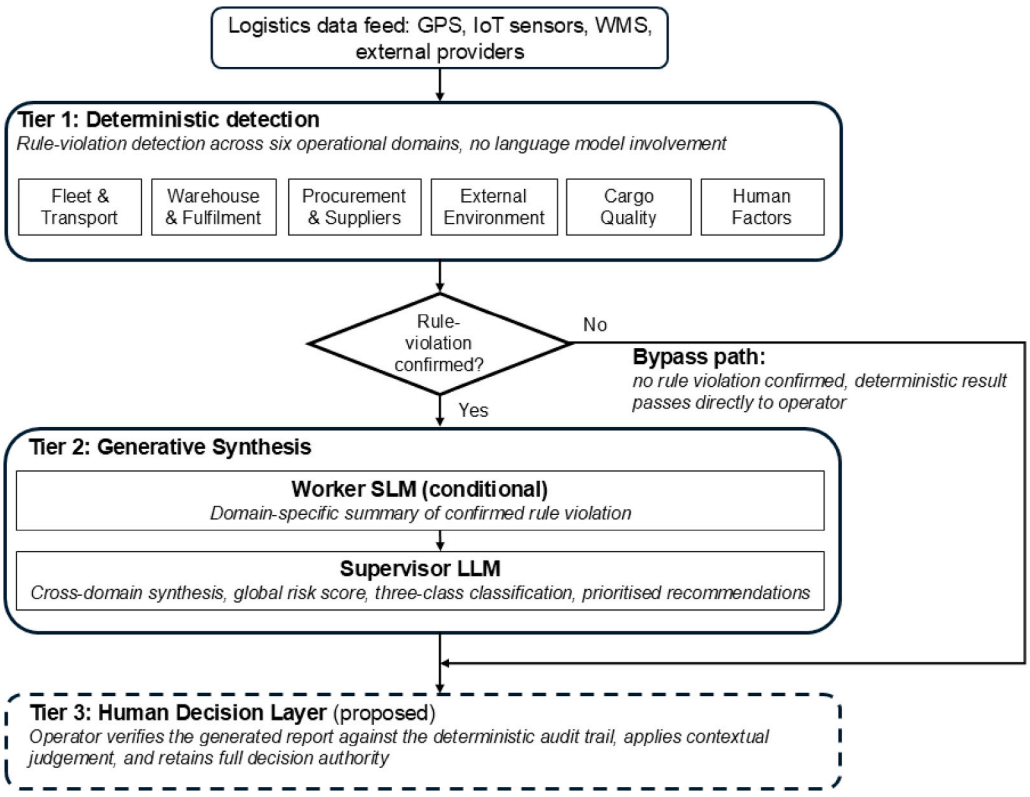


Fig. 3. Architecture of the proposed framework with N = 6 specialised worker agents covering the operational domains of a multimodal logistics network. Each agent applies domain-specific detection rules (Table 2) to its assigned subset of input features. A deterministic rule engine at Tier 1 detects rule violations without language model involvement; worker SLMs at Tier 2 are called only when an rule violation is confirmed to generate natural language summaries. The supervisor LLM aggregates worker outputs into a global risk score and prioritised recommendations for the human operator (Tier 3, proposed). The architecture generalises to any N specialised agents; the tier separation and conditional activation are independent of the number or type of domains.

Table 2

Summary of specialist worker agent configurations. Weights (w_i) reflect the relative operational importance of each domain, subject to $\sum_{i=1}^{6} w_i = 1$.

Agent Domain	Key Input Features	w_i	Threshold Rules (Tier 1)
Fleet & Transport	vehicle_gps_latitude, vehicle_gps_longitude, fuel_consumption_rate, eta_variation_hours, route_risk_level	0.25	Fuel $\notin [5,20]$L/hr; ETA var $\notin [-2,5]$ hrs; Route risk > 7; GPS lat/lon $= 0$
Warehouse & Fulfilment	warehouse_inventory_level, loading_unloading_time, handling_equipment_availability, order_fulfillment_status	0.10	Inventory < 100 units; Load time > 4 hrs; Equip availability $= 0$; Fulfilment $= 0$
Procurement & Suppliers	supplier_reliability_score, lead_time_days, historical_demand, shipping_costs	0.10	Reliability < 0.6; Lead time > 10 days; Demand > 5000 units; Shipping cost $> \$700$
External Environment	traffic_congestion_level, weather_condition_severity, port_congestion_level, customs_clearance_time	0.15	Traffic > 7; Weather severity > 0.7; Port congestion > 8; Customs time > 4 hrs
Cargo Quality	iot_temperature, cargo_condition_status	0.15	IoT Temp $\notin [-5,25]\,^{\circ}$C; Cargo condition $= 0$
Human Factors	driver_behavior_score, fatigue_monitoring_score	0.25	Driver behaviour < 0.5; Fatigue score > 0.7

could act upon. A unified risk score conflating cargo temperature, route hazard level, and driver fatigue cannot be meaningfully escalated to a single responsible actor, as it belongs simultaneously to quality assurance, fleet management, and human factors. Separating these concerns into discrete agents keeps each output interpretable and specific to its corresponding organisational domain.

Domain boundaries and rule-violation detection thresholds were defined by the research team through examination of the dataset's feature space, grouping features according to their functional role within the logistics network and drawing on domain knowledge alongside the distributional properties observed during data characterisation. The decomposition and threshold values are researcher-defined and context-specific; we did not identify published sources prescribing these specific values, and practitioners deploying the framework in different settings should treat these as configurable starting points rather than fixed parameters. Table 2 details the six agents, their assigned features, risk weights, and thresholds rules.

The decomposition can be related to the risk classification of Ho et al. [42], who divide supply chain risk into macro-risks and micro-risks, the latter comprising demand, manufacturing, supply, and infrastructural (information, transportation, and financial) risk, and who define supply chain risk management as the endeavour to identify, evaluate, mitigate, and monitor events that might adversely impact any part of a supply chain. On this reading, Fleet & Transport corresponds to transportation risk, Procurement & Suppliers to supply risk, External Environment to macro-risk, and Warehouse & Fulfilment and Cargo Quality to internal operational risk affecting the timeliness and quality of goods handling. The correspondence is our construction rather than a mapping asserted by that source, and it is partial: Human Factors has no direct counterpart in the taxonomy, and demand and financial risks fall outside the operational monitoring scope of this framework. At the process level, Tier 1 performs identification and evaluation through rule violations and

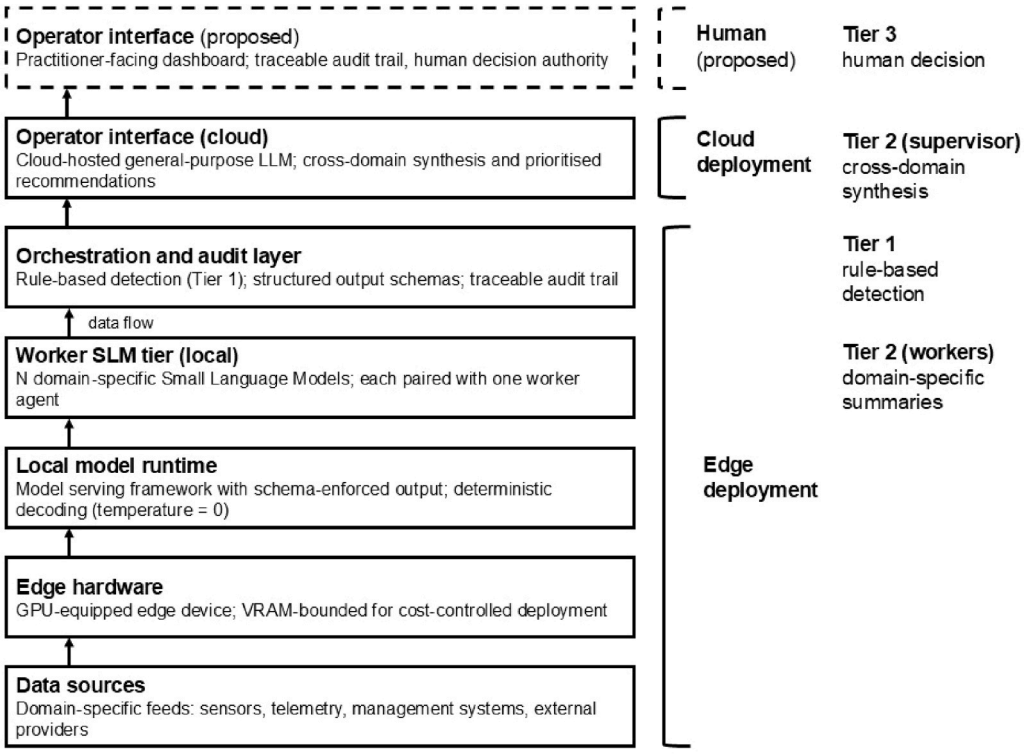


Fig. 4. Deployment stack of the proposed framework. Data sources at the base feed through seven layers to the operator interface at the top, with data flowing upward. Tier 1 (rule-based detection) is implemented in the orchestration and audit layer; Tier 2 is split across the local Worker SLM tier and the cloud-hosted Supervisor LLM tier; Tier 3 (human decision) is proposed as the operator interface layer. The edge deployment bracket indicates the five layers executed on local edge-representative hardware; the cloud deployment bracket indicates the single layer executed via API. This layered view generalises across application domains: the stack remains constant whilst the data sources and agent SLMs vary per application.

composite scoring, Tiers 2 and 3 prepare prioritised information for the mitigation decisions that remain with the human operator, and the pipeline as a whole operationalises continuous monitoring. The three level risk banding (Low, Moderate, High at 0.3 and 0.7) follows a banding convention in common use rather than externally validated cut-offs, and the sensitivity of the risk-level distribution to these choices is examined directly in Section 4.9.

In-transit monitoring falls to the **Fleet & Transport** agent, which flags fuel consumption outside 5–20 L/hr, ETA variations beyond −2 to + 5 h, route risk levels exceeding 7 on a 10-point scale, or GPS coordinates defaulting to zero. Output is directed at a logistics coordinator, translating a near-maximum ETA deviation into a scheduling decision requiring dock rescheduling or customer communication.

At the distribution centre, the **Warehouse & Fulfilment** agent triggers when inventory drops below 100 units, loading or unloading times exceed 4 h, or equipment availability and order fulfilment statuses register as completely failed. Anomalies here are often causally linked, with equipment unavailability frequently preceding loading delays, and outputs are structured to help operations managers distinguish isolated incidents from cascading facility failures.

Upstream conditions that may manifest as future stockouts or cost overruns are tracked by the **Procurement & Suppliers** agent, which activates when supplier reliability falls below 0.6, lead times exceed 10 days, shipping costs surpass $700, or historical demand spikes above 5000 units. Output is presented in natural language for category managers. A low reliability score is flagged as a risk to supply continuity, prompting a safety stock review.

Disruptions originating outside the shipper's direct control are handled by the **External Environment** agent, which flags traffic or port congestion above 7 and 8 respectively, weather severity above 0.7, or customs clearance times exceeding 4 h. Output focuses on contingency options available to the operator, such as modal substitutions or force-majeure communications.

Although the **Cargo Quality** agent covers a small feature set, decision stakes are high: product integrity failures carry regulatory and liability consequences. Any IoT temperature reading outside −5°C to 25°C, or any binary indication of cargo spoilage, generates a maximum risk score. Output directs quality assurance personnel towards compliance and insurance escalation procedures.

Finally, the **Human Factors** agent monitors variables that data-driven frameworks often overlook, flagging driver behaviour scores below 0.5 or fatigue monitoring scores above 0.7. A low behaviour score is a leading indicator for accident liability; a high fatigue score represents an active physiological hazard. Output is directed at safety officers, presenting findings as grounds for mandated rest stops or route reassignments.

3.2.3. *Supervisor agent and global risk synthesis*

Structured outputs from the six worker agents are aggregated by the supervisor agent, which computes a weighted global risk score, S_{global}:

$$S_{global} = \sum_{i=1}^{N} w_i . s_i \tag{1}$$

where $N = 6$ is the number of worker agents, $s_i \in [0, 1]$ is the normalised risk score reported by agent i, and w_i is the predefined weight for domain i, subject to $\sum_{i=1}^{N} w_i = 1$.

Weights were established through expert consensus on the research team. Fleet & Transport and Human Factors share the highest priority ($w_1 = 0.25, w_6 = 0.25$), reflecting the assessment that transit operations and workforce signals represent the most severe immediate safety risks. Cargo Quality and External Environment are assigned secondary weights ($w_5 = 0.15, w_4 = 0.15$) given the potential for irreversible product spoilage and uncontrollable exogenous delays. Warehouse &

Fulfilment and Procurement & Suppliers are each weighted at 0.10, representing critical but slower-moving risk domains. All weights are held in the system configuration to allow rapid adjustment without modifying the pipeline. Formal elicitation involving a broader practitioner panel, such as the analytic hierarchy process [43], would strengthen these assignments but falls outside the scope of this study.

Once computed, S_{global} is mapped to a three-class classification: Low risk ($S_{\text{global}} < 0.3$), Moderate risk ($0.3 \leq S_{\text{global}} \leq 0.7$), or High risk ($S_{\text{global}} > 0.7$). GPT-3.5-Turbo then synthesises the domain narratives, global score, and risk classification into a structured JSON output containing prioritised recommendations. Conversational text is expressly forbidden in the prompt instructions (the "negative constraint"), and time-sensitive actions are prefixed with the exact string "IMMEDIATE:'' to support rapid operator prioritisation. These constraints reduce output variability and keep every recommendation traceable to its originating deterministic findings. Empirical performance is reported in Section 4.

3.2.4. Candidate model selection and screening protocol

Eight candidate SLM architectures spanning 1–7 billion parameters[1] are screened in Phase 1 to identify which models best serve the explanation layer. The candidate pool covers three architectural paradigms: dense Transformers, Mixture-of-Experts (MoE) models, and hybrid architectures combining Transformer attention with state space model components. Candidates are organised into three scale groups: Sub-2B (evaluated at fp16 to isolate architectural limitations from quantisation effects), Mid-Range (3B to 4B parameters), and Large (7B parameters). Full specifications are provided in Table 4.

Screening is conducted across two pilot batches ($n = 30$ and $n = 70$) drawn from the first 100 records of the dataset, evaluating each candidate across three dimensions:

1. **System Reliability:** strict JSON schema adherence (the proportion of invocations producing valid structured output) and negative constraint adherence (conformance to the prompt-level prohibition on conversational text).
2. **Reasoning Quality:** agent summary cosine similarity (semantic similarity of worker-level summaries against the GPT-4o-mini cloud reference), and supervisor recommendation cosine similarity (semantic similarity of supervisor-level outputs against the GPT-4o-mini cloud reference). Flag detection is deterministic and identical across models by construction, so it is reported as a structural property rather than per-model metric.
3. **Edge Efficiency:** average generation speed (tokens per second), peak VRAM consumption (gigabytes), and total pipeline processing time (seconds).

GPT-4o-mini serves as the cloud reference baseline for Phase 1 screening, selected because Phase 1 requires a cost-efficient reference that can be called repeatedly across two screening batches and eight candidates. GPT-4o, used for Phase 2 ground truth construction, is reserved for the smaller and more resource-intensive formal evaluation. GPT-4o-mini is not considered a local deployment candidate.

Three elimination criteria are applied. Any model that fails to sustain JSON schema adherence above 60% across both batches is disqualified on reliability grounds. Any model recording average generation speed below 35 TPS is eliminated on throughput grounds, as this falls below the minimum operational threshold for the target deployment environment; an exception may be granted at the research team's judgement if a candidate demonstrates reasoning quality that substantially exceeds all other candidates in its scale group. Within each scale group, a model that is Pareto-dominated on both throughput and reasoning quality by another candidate in the same group is eliminated. The top three performing candidates and one lower-bound reference model (retained to represent the dense Transformer performance floor) advance to Phase 2.

3.3. Evaluation framework

Two questions guide the evaluation: whether the automated pipeline is computationally feasible under edge hardware constraints, and whether the generative outputs faithfully preserve the rule violation findings produced by the deterministic tier.

3.3.1. Three-phase experimental design

Phase 1 screens the eight candidates against pilot data batches to identify finalists; the candidate pool, screening dimensions, and elimination criteria are set out in Section 3.2.4.

Phase 2 constitutes the primary empirical basis for all quantitative claims in this study. A validation set of 100 records is randomly sampled from the remaining population (row indices 101–32,065) using a fixed random seed (RANDOM_SEED = 42) whilst maintaining chronological index sorting. Ground truth narratives were initially constructed manually by the research team during early experimental iterations. Manual construction proved impractical to scale consistently across 100 records, and the process was replaced by prompting GPT-4o with the structured deterministic output for each record. GPT-4o was selected for this role after preliminary testing indicated that candidate SLMs did not produce outputs of sufficient quality for reference use. Fan et al. [44] report strong alignment between GPT generated supply chain risk summaries and human benchmarks, which supported this design choice at the time; a subsequent direct check in this study found that the GPT-4o references nonetheless introduce content beyond the deterministically flagged reasons in most multi-domain records (Section 5.5), so that external result should not be read as validation of the specific references used here. Two members of the research team reviewed each GPT-4o-generated narrative for operational plausibility.

Phase 3 serves as a scale and cost validation stage, evaluating operational scalability over $n = 1,000$ records sampled uniformly at random from the main dataset using the same fixed-seed protocol (RANDOM_SEED = 42). Rather than ground truth evaluation, this phase benchmarks two pipeline configurations: a fully cloud-based configuration deploying GPT-4o-mini for all worker agents and GPT-3.5-Turbo for the supervisor, and the proposed hybrid configuration deploying a localised SLM for worker agents alongside GPT-3.5-Turbo for the supervisor. End-to-end pipeline latency and token-based API cost are recorded to assess whether the hybrid approach offers a practical efficiency advantage over a fully cloud-based alternative.

The division of evidence across phases bounds what the study can claim: output quality, information preservation, and fidelity to the flags rest on the 100-record validation set and are reported as preliminary, while the 1,000-record set supports only the processing-time and cost comparison and involves no ground-truth evaluation.

3.3.2. Hardware and inference environment

All local SLM experiments run on a hardware environment with Ubuntu 24.04 via Windows Subsystem for Linux 2 (WSL2), equipped with a Ryzen 9 5900HX processor, 16 GB of DDR4 RAM, a 1 TB NVMe SSD, and an RTX 3060 Mobile GPU limited to 6 GB of GDDR6 VRAM. Inference runs through Ollama, an open-source local model runtime, with strict JSON schema enforcement. Decoding temperature is set to zero: at this setting the model selects the highest-probability next token at each step, producing deterministic output for a given input and

[1] The upper bound of the SLM category is not settled in the literature. Lu et al. [40] define SLMs as 100 M to 5B parameters, noting that 7B models are still predominantly cloud-deployed. Belcak et al. [20] adopt a deployment-centric definition under which any model executable on consumer-grade hardware qualifies, noting that "*as of 2025, we would be comfortable with considering most models below 10bn parameters in size to be SLMs*." The two 7B candidates (Mistral 7B and Granite 4 Tiny-H) are included here because both fit within the 6 GB VRAM constraint after quantisation or sparse routing, satisfying the edge-deployment criterion central to this study.

eliminating sampling variance as a source of experimental noise. This follows the same zero-temperature protocol adopted by Zhao et al. [19] for LLM-based supply chain risk identification.

3.3.3. Quantitative performance metrics

Exact-Match Divergence Analysis. A divergence is recorded at the agent-domain level when the generated output fails to exactly match the ground truth narrative for that domain. Each evaluated record can contain between zero and six agent divergences, depending on how many domains were activated. This analysis identifies whether a divergence has occurred; the failure taxonomy defined below then classifies the type of divergence.

Urgency Flag Preservation Check. Whether the generative supervisor tier preserves the ''IMMEDIATE:'' flag produced at the deterministic tier is tracked by this metric. Occurrences of the flag are counted in both the ground truth and the predicted recommendations, and any mismatch is recorded. Interpretation is asymmetric because the two error directions carry unequal operational cost: a false negative, in which the model fails to generate a required IMMEDIATE: flag, represents a missed urgent alert and is the more critical failure mode; a false positive generates an unnecessary alert but does not suppress a genuine one.

Risk-Weighted Scenario Fidelity (RWSF) is proposed as an exploratory indicator for this implementation rather than a validated metric; its limited discriminating power in this study is discussed in Section 5.5. Raw divergence counts treat every agent-domain failure as equally important, an inadequate basis for operational decision support. A model that drops warehouse details ($w = 0.10$) produces a materially different operational outcome from one that drops fleet or human factors information ($w = 0.25$ each), even if both register the same raw failure count. RWSF captures this distinction by weighting each agent-domain divergence by the operational risk weight w_i from Table 2:

$$\mathrm{RWSF}_{\mathrm{row}} = 1 - \sum_{i=1}^{N} (w_i \cdot \Delta_i) \tag{2}$$

where $N = 6$ is the number of agent domains, w_i is the operational risk weight for domain i (Table 2), and $\Delta_i = 1$ if the model output diverges from the ground truth for domain i in that row, and 0 otherwise. Per-row scores are averaged across all 100 evaluated rows, including rows with no divergences (which contribute $\mathrm{RWSF}_{\mathrm{row}} = 1$), to produce a model-level result bounded in $[0, 1]$.

A brief interpretation guide is needed to read the reported values correctly, since the metric behaves differently from an accuracy percentage. Under exact-match divergence analysis, if a compression event during summarisation causes multiple activated agent domains in the same row to register as divergent simultaneously rather than individually, the per-row weighted penalty approaches the sum of the activated weights rather than the weight of a single domain. In such cases, failed rows would contribute little residual fidelity, and the model-level RWSF would be driven primarily by the proportion of clean rows. A model-level RWSF of 0.25, for instance, would not mean the model is 25 per cent accurate; it would reflect the average residual fidelity across all rows after each failed row is penalised for its specific combination of diverging domains.

Table 3 illustrates this with a simplified four-row example using the domain weights from Table 2.

Row B: Five domains diverge (Fleet, Warehouse, External, Cargo, Human); penalty $= 0.25 + 0.10 + 0.15 + 0.15 + 0.25 = 0.90$; $\mathrm{RWSF}_{\mathrm{row}} = 1 - 0.90 = 0.100$.

Row D: Three domains diverge (Warehouse, Procurement, External); penalty $= 0.10 + 0.10 + 0.15 = 0.35$; $\mathrm{RWSF}_{\mathrm{row}} = 1 - 0.35 = 0.650$.

Model average: $(1.000 + 0.100 + 1.000 + 0.650)/4 = 0.688$.

Row D has fewer failures than Row B (3 versus 5), but the comparison is more instructive than the counts suggest: Row D's failures are concentrated in low-weight domains (total weight 0.35), whilst the high-weight domains (Fleet, Human) are preserved. RWSF captures this distinction; raw failure counts do not.

RWSF is most informative when read alongside the raw failure rate, since the two metrics answer complementary questions. The raw failure rate indicates how often a model fails; RWSF indicates how severely the failures weigh when they occur. A model with a higher RWSF at a given raw failure rate concentrates its failures in lower-weight domains, leaving the higher-weight domains such as Fleet & Transport and Human Factors relatively intact. For operational decision support, this is the preferable failure distribution.

The Edge Efficiency Index (EEI) is introduced as an implementation-specific exploratory indicator. In Industrial Internet of Things (IIoT) deployments, computational resources sit at the network periphery, adjacent to sensors, vehicles, or warehouse machinery. This edge positioning supports low-latency processing, preserves operational privacy, and reduces cloud bandwidth dependency [45,46]. Evaluating language models for these constrained environments requires metrics that reflect operational conditions rather than laboratory benchmarking.

Standard efficiency metrics such as raw Tokens-Per-Second (TPS) assume a linear utility of inference speed. In a supply chain monitoring context, however, inference speed exhibits diminishing returns: a model must achieve a baseline throughput to match the decision cycle of a human operator, but generating at 200 TPS offers no practical advantage over 100 TPS when downstream decisions operate on a scale of minutes. Linear metrics disproportionately reward hyper-fast, heavily quantised models even when their reasoning fidelity degrades, which is problematic in risk-sensitive applications.

We introduce a composite metric that balances reasoning accuracy against edge hardware constraints:

$$EEI = \left(\frac{RWSF_{row}}{\mathrm{VRAM}_{\mathrm{peak_GB}}} \right) \times \log_{10}(\mathrm{TPS}) \tag{3}$$

where $\mathrm{RWSF}_{\mathrm{row}}$ is the model's Risk-Weighted Scenario Fidelity Score (Eq. 2), $\mathrm{VRAM}_{\mathrm{peak_GB}}$ denotes peak VRAM consumption in gigabytes, and TPS denotes average generation speed. $\mathrm{RWSF}_{\mathrm{row}}$ is divided by VRAM rather than subtracted so that the penalty scales proportionally with memory consumption: a model using twice the VRAM receives half the score regardless of its fidelity. The resulting ratio is multiplied by $\log_{10}(\mathrm{TPS})$ rather than added, so that a model failing to meet minimum throughput requirements collapses the entire score towards zero, reflecting that an operationally unviable model cannot compensate through fidelity or memory efficiency alone. Logarithmic scaling of TPS further ensures that achieving baseline operational throughput yields a meaningful score contribution, whilst gains beyond that threshold

Table 3
Worked example of RWSF calculation over four rows. Weights from Table 2: Fleet = 0.25, Warehouse = 0.10, Procurement = 0.10, External = 0.15, Cargo = 0.15, Human = 0.25. ✓ = match, × = divergence.

Row	Fleet	Whse	Proc.	Ext.	Cargo	Human	$\mathrm{RWSF}_{\mathrm{row}}$
	(0.25)	(0.10)	(0.10)	(0.15)	(0.15)	(0.25)	
A (clean)	✓	✓	✓	✓	✓	✓	1.000
B (failed)	×	×	✓	×	×	×	0.100
C (clean)	✓	✓	✓	✓	✓	✓	1.000
D (failed)	✓	×	×	×	✓	✓	0.650

contribute progressively less.

Because EEI relies on the RWSF score established during exact-match divergence analysis, it is computed exclusively during the Phase 2 primary benchmark to formalise the final deployment candidate selection.

Standard agent-level metrics are also recorded, including Mean Absolute Error for risk score variance.

3.3.4. Semantic and qualitative analysis

Semantic similarity between generated outputs and ground truth narratives is computed using the nomic-embed-text:137m-v1.5-fp16 embedding model, which calculates cosine similarity between the embedding vectors of the generated and reference texts. Its low parameter footprint makes it compatible with edge deployment alongside the primary SLMs.

Divergence is assessed at the supervisor level: a report diverges when its count of IMMEDIATE -tagged recommendations differs from the reference count for the same record. Reports that fail this check are routed to GPT-4o-mini, which classifies each failure using a zero-shot prompt naming one of three mutually exclusive error categories, defined by the research team (the operative prompt text is provided as supplementary material):

1. **Compression Loss:** The pipeline's recommendations dropped critical items present in the reference report, most often omitting a recommended action for a flagged risk entirely, so that fewer IMMEDIATE-tagged actions were produced.
2. **Lexical Deviation:** The recommendations covered the flagged risks but phrased them in weaker or more passive language than the reference, so that the urgency marking (the IMMEDIATE prefix) was not carried. The underlying risks are addressed; the phrasing and urgency signalling deviate.
3. **Hallucination:** The recommendations contained content judged to have been invented, or nonsensical advice, relative to the reference report.

Two members of the research team review each classification for plausibility. The reliance on an automated judge for failure classification is acknowledged as a limitation in Section 5.5. The prompts used for reference-narrative generation and failure classification, together with worked examples pairing a reference narrative, a model output, and the assigned category, are provided as supplementary material. Given the reliance on LLM-generated references throughout, this procedure constitutes an LLM-assisted consistency assessment rather than an independent validation of explanation quality.

4. Results and analysis

4.1. Phase 1: Candidate screening

Phase 1 evaluated the eight candidates described in Table 4 across three dimensions: System Reliability (structured output compliance), Reasoning Quality (semantic similarity against the GPT-4o-mini cloud reference), and Edge Efficiency (throughput and VRAM consumption within the 6 GB hardware constraint). GPT-4o-mini served as the cloud reference baseline and was not considered a local deployment candidate. Results for the $n = 30$ and $n = 70$ batches are reported in Tables 5 and 6 respectively.

Both Sub-2B models were eliminated, though on different grounds. Llama 3.2 1B and SmolLM2 1.7B recorded the lowest Agent Summary Similarity scores at $n = 70$ (0.676 and 0.689 respectively), consistent with the capacity ceiling of sub-2B dense architectures at fp16 precision. SmolLM2 additionally failed to sustain JSON adherence above 55% across both batches, disqualifying it on reliability grounds independently of its reasoning quality. Llama 3.2 1B was retained as the lower-bound reference representing the dense Transformer performance floor.

Among the Mid-Range and Large candidates, Phi-3.5 Mini and Mistral 7B fell below the minimum operational throughput threshold at 26.2 TPS and 31.2 TPS respectively. Qwen3 4B (30.2 TPS) also fell marginally below this threshold during the $n = 70$ batch but was retained as a researcher judgement: its Supervisor Recommendation Similarity (0.921) was the highest of any local candidate, and the research team judged that this reasoning advantage warranted evaluation at full scale in Phase 2 despite the marginal throughput shortfall. Gemma3 4B was outperformed on both throughput and reasoning quality, operating at less than half the speed of Granite 3.1 MoE (51.9 versus 115.0 TPS) whilst falling short of Qwen3 4B on reasoning metrics. Granite 3.1 MoE, Qwen3 4B, and Granite 4 Tiny-H, together with Llama 3.2 1B as the lower-bound reference, progressed to Phase 2.

Table 4
Candidate model specifications: architecture type, parameter scale, context window, quantisation method, and model size. Models are ordered by parameter count. GPT-4o-mini serves as the cloud reference baseline throughout both screening batches.

Model	Scale Group	Params (B)	Context (K)	Quant.	Size (GB)
GPT-4o-mini	Cloud Reference	–	128	–	–
llama3.2:1b	Sub-2B	1.0	128	fp16	2.5
smollm2:1.7b	Sub-2B	1.7	8	fp16	3.4
granite3.1-moe:3b	Mid-Range	3.0	128	q8_0	3.5
phi3.5:3.8b-mini	Mid-Range	3.8	4	q8_0	4.1
gemma3:4b-it	Mid-Range	4.0	128	QAT	4.0
qwen3:4b	Mid-Range	4.0	256	q8_0	4.3
granite4:tiny-h	Large (7B)	7*	1000	–	4.2
mistral:7b	Large (7B)	7.0	32	q4_K_M	4.4

Quantisation notation: fp16 = 16-bit floating-point weights (near-lossless, highest memory cost); q8_0 = 8-bit integer quantisation; q4_K_M = 4-bit mixed quantisation (lowest memory cost, some precision trade-off); QAT = Quantisation-Aware Training.
* Granite 4 Tiny-H reports 7B nominal parameters; effective active parameters per forward pass are substantially lower due to sparse MoE and SSM routing.

4.2. Finalist model architectures

Before presenting the Phase 2 results, the architectural differences between the four finalists require characterisation, as these differences are central to interpreting the performance divergence that follows. The four models span three distinct computational paradigms: Llama 3.2 1B (a dense Transformer retained as the lower-bound reference), Granite 3.1 MoE 3B, Qwen3 4B, and Granite 4 Tiny-H.

Llama 3.2 1B is a canonical dense Transformer in which every parameter activates for every token, using no routing or conditional computation. Its design includes Grouped Query Attention (GQA), where multiple query heads share a single key-value head to reduce the KV cache footprint whilst preserving lossless fidelity. Rotary Position Embeddings (RoPE) support generalisation beyond training sequence lengths, and SwiGLU (Swish-Gated Linear Unit) activations in the feedforward sublayers improve parameter efficiency relative to standard ReLU activations. Full dense attention, however, imposes the complete computational and memory cost on every forward pass.

Granite 3.1 MoE 3B introduces Top-8 Expert Routing at the feedforward sublayers. A lightweight gating network selects the top-8 expert sub-networks per token, leaving the remaining experts dormant, decoupling total model capacity from per-token compute cost. Combined with q8_0 quantisation, this reduces weight-loading memory bandwidth pressure. Granite 3.1 retains the standard Transformer attention mechanism with a full, lossless KV cache. Token representations are computed using standard scaled dot-product attention [47] as shown in Eq. 4.

Table 5
Screening results, Phase 1 Batch 1 ($n = 30$): system reliability, reasoning quality, and edge efficiency across all eight candidate models and the GPT-4o-mini cloud reference baseline. All models processed 30 supervisor invocations and 124 worker invocations (154 total LLM calls).

Model	Reliability		Reasoning Quality		Edge Efficiency		
	JSON (%)	NC Adh. (%)	Smry.Sim	Rec.Sim	AGS (TPS)	VRAM (GB)	Time (s)
GPT-4o-mini	100.0	100.0	0.918	0.912	38.4	–	232
llama3.2:1b	91.6	100.0	0.658	0.894	74.3	4.40	125
smollm2:1.7b	53.3	100.0	0.690	0.886	58.8	4.99	270
granite3.1-moe:3b	100.0	100.0	0.859	0.898	115.3	5.27	108
phi3.5:3.8b-mini	100.0	100.0	0.876	0.907	26.1	5.31	535
gemma3:4b-it	100.0	100.0	0.892	0.908	50.4	5.23	478
qwen3:4b	100.0	100.0	0.894	0.922	33.3	5.30	366
granite4:tiny-h	100.0	100.0	0.884	0.899	46.9	5.34	296
mistral:7b	100.0	100.0	0.904	0.901	31.2	5.25	406

Column headers: JSON = Strict JSON schema adherence; NC Adh. = Negative constraint adherence (conformance to the prompt-level prohibition on conversational text); Smry.Sim = Agent summary cosine similarity (vs. GPT-4o-mini); Rec.Sim = Supervisor recommendation cosine similarity (vs. GPT-4o-mini); AGS = Average generation speed; TPS = tokens per second; VRAM = Peak memory usage (GB); Time = Total pipeline processing time (seconds). Flag reproduction against the deterministic rule is 100% for every model by construction and is therefore not reported as a per-model column.

Table 6
Screening results, Phase 1 Batch 2 ($n = 70$): system reliability, reasoning quality, and edge efficiency across all eight candidate models and the GPT-4o-mini cloud reference baseline. All models processed 70 supervisor invocations and 294 worker invocations (364 total LLM calls). Finalist models advancing to Phase 2 (Section 4.3) are highlighted in **bold**.

Model	Reliability		Reasoning Quality		Edge Efficiency		
	JSON (%)	NC Adh. (%)	Smry.Sim	Rec.Sim	TPS	VRAM (GB)	Time (s)
GPT-4o-mini	100.0	100.0	0.923	0.912	40.1	–	529
llama3.2:1b*	84.3	100.0	0.676	0.884	74.9	4.40	271
smollm2:1.7b	54.1	100.0	0.689	0.889	58.6	5.18	599
granite3.1-moe:3b	**100.0**	**100.0**	**0.856**	**0.889**	**115.0**	**5.30**	**246**
phi3.5:3.8b-mini‡	100.0	100.0	0.876	0.912	26.2	5.21	1242
gemma3:4b-it‡	100.0	100.0	0.896	0.907	51.9	5.24	1074
qwen3:4b	**100.0**	**100.0**	**0.895**	**0.921**	**30.2**	**5.46**	**958**
granite4:tiny-h	**100.0**	**100.0**	**0.875**	**0.893**	**48.6**	**5.32**	**697**
mistral:7b‡	100.0	100.0	0.907	0.902	31.2	5.25	967

Column headers as for Table 5.
* Llama 3.2 1B retained as the lower-bound reference (dense Transformer performance floor); eliminated as a performance candidate on Smry.Sim grounds (> SD below the GPT-4o-mini cloud reference).
‡ Eliminated candidates: Phi-3.5 Mini (26.2 TPS) and Mistral 7B (31.2 TPS) fall below the minimum operational throughput threshold. Gemma3 4B is Pareto-dominated within the mid-range tier; it is less than half the speed of Granite 3.1 MoE whilst exhibiting lower reasoning quality than Qwen3 4B.
Flag reproduction against the deterministic rule is 100% for every model by construction and is therefore not reported as a per-model column.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where Q, K, and V are the query, key, and value representation matrices respectively, and d_k represents the dimensionality of the key vectors. The matrix product QK^T computes the unnormalised similarity scores between tokens, the scaling factor $\frac{1}{\sqrt{d_k}}$ stabilises gradients during the softmax probability normalisation, and the final multiplication with V yields the context-weighted output. Because every prior token's key (K) and value (V) projections must be stored in the KV cache at full floating-point fidelity to continuously evaluate this equation, this lossless retrieval mechanism guarantees that any prior position can be retrieved without approximation, dictating an $O(n)$ memory footprint that preserves exact numeric values. Expert routing addresses the feedforward compute bottleneck whilst leaving the attention memory architecture unchanged.

Qwen3 4B interleaves fully dense attention layers with MoE feedforward blocks. Dense attention is preserved in the early and terminal layers, where token representations are first constructed and contextualised, a stage sensitive to sparsification. MoE routing is applied in the middle feedforward sublayers, where expert specialisation provides efficiency gains without compromising representational fidelity. A runtime-configurable thinking mode for chain-of-thought reasoning was disabled during evaluation to maintain consistent inference conditions across all candidates. Like Granite 3.1, Qwen3's attention sublayers maintain a full, lossless KV cache, preserving $O(n)$ memory scaling with sequence length.

Granite 4 Tiny-H combines three paradigms within a single inference stack: standard Transformer attention layers, Mamba-2 Selective State Space Model (SSM) layers, and sparse MoE feedforward routing. The Mamba-2 component introduces a different memory contract. Rather than storing every prior token's key and value projections verbatim, each SSM layer maintains a fixed-size hidden state updated through selective recurrence:

$$h_t = A_t \cdot h_{t-1} + B_t \cdot x_t \tag{5}$$

$$y_t = C_t \cdot h_t \tag{6}$$

where x_t is the input token representation at time step t, y_t is the projected output, and h_t is the updated hidden state vector. The matrices A_t, B_t, and C_t are input-dependent, discretised parameters computed dynamically from the current token: A_t governs retention or decay of historical state information, B_t regulates integration of new input, and C_t projects the compressed hidden state into the output space.

This state has an $O(1)$ memory footprint with respect to sequence length, independent of context size, in contrast to the $O(n)$ scaling of the KV cache. The mechanism is inherently lossy: the model learns a compression policy that captures the semantic content of prior context at constant computational cost but cannot guarantee verbatim retention of any specific prior token. The distinction between lossless KV retention (Granite 3.1) and lossy state compression (Granite 4) is the central variable underlying their divergence in the Phase 2 results.

4.3. Phase 2: Primary benchmark ($n = 100$)

The four finalist models were evaluated on 100 held-out records (Section 3.3). Conditional activation across these records produced 435 worker-level SLM invocations. Exact-match divergence analysis at the supervisor level, comparing the count of IMMEDIATE-tagged recommendations in each pipeline's report against the reference report for the same record, yielded 291 divergent reports out of 400 model-record comparisons; each was classified into the three-category failure taxonomy (Section 3.3) by GPT-4o-mini. Complete results are reported in Tables 7 and 8.

Table 8 yields three observations. Schema-enforced structured output generation via Ollama achieved full format compliance for the three mid-range candidates, with 100% JSON adherence and 100% negative constraint adherence across all 100 records. Llama 3.2 1B recorded 87.3% JSON adherence, consistent with the degradation pattern observed across both Phase 1 batches. Processing time, however, reveals a throughput gap that the similarity scores mask: Granite 3.1 MoE completes the 100-record pipeline in 380 s, whilst Qwen3 4B requires 1226 seconds and Granite 4 Tiny-H requires 1006 seconds, a 3.2x and 2.6x latency difference respectively. For a deployment context where pipeline latency affects operational response time, this gap is substantial. Similarity scores across all four models occupy a narrow range (Agent Summary Similarity: 0.671–0.895; Supervisor Recommendation Similarity: 0.887–0.924), confirming that raw semantic similarity alone cannot discriminate between candidates and motivating the RWSF and failure taxonomy analysis in the sections that follow.

4.4. Aggregate results and failure taxonomy

Raw divergence failure rates and RWSF scores are reported in Figs. 5 and 6 and Table 9. Failure rates show a moderate spread: Qwen3 4B records the lowest (0.69), followed by Granite 3.1 MoE and Llama 3.2 1B (both 0.73), and Granite 4 Tiny-H the highest (0.76). The corresponding RWSF scores compress into a narrow band, with all four models scoring between 0.29 and 0.31.

Read alongside the clean-row rate (1 − raw failure rate), the RWSF scores reveal how failures distribute across the risk spectrum. When a model's RWSF lies close to its clean-row rate, most of the activated weight in failed rows is being penalised, leaving little residual fidelity to raise the mean above the clean-row baseline. Three models exhibit this pattern: Granite 3.1 MoE (clean-row rate 0.27, RWSF 0.289), Llama 3.2 1B (0.27 and 0.287), and Qwen3 4B (0.31 and 0.302). The near-zero gaps are consistent with the exact-match test registering most activated domains in a failed row as divergent once compression occurs. Granite 4 Tiny-H (clean-row rate 0.24, RWSF 0.309) is the only model where this gap exceeds the near-zero pattern, with a difference of 0.07 suggesting that its failed rows retain more weight in higher-risk domains than the other three architectures. This observation is consistent with the SSM selective-compression prediction from Section 4.2, but the difference has not been tested for statistical significance and could reflect sampling variation at $n = 100$. A per-domain failure breakdown, which is not computed in this study, would be needed to confirm whether the retained weight is indeed concentrated in specific high-priority domains. The architectural basis for this observation is discussed further in Section 4.6.

These near-zero gaps also explain why the raw failure rate alone cannot determine the operational deployment ranking: three models with similar raw failure rates (0.69–0.76) exhibit near-identical weighted fidelity because their failures, when they occur, are distributionally similar across the risk spectrum. Fine-grained ranking must therefore draw on independent criteria such as throughput, format reliability, and failure-type breakdown rather than RWSF alone.

Among the outputs that diverged from the GPT-4o reference, the classifier assigned none to the fabrication category, across all four models, under schema-enforced structured output at temperature = 0, on a task confined to summarising rule-confirmed findings. Divergent reports dropped items carried by the reference (Compression Loss) or phrased them with weaker urgency (Lexical Deviation) rather than inventing content. A direct fidelity-to-flags check against a deterministic template baseline follows in Section 4.5. Generalisation beyond this dataset and parameter range requires further validation.

Compression Loss accounts for 252 of the 291 classified failures (86.6%), making it the dominant failure mode across all four models. Counts range narrowly from 60 to 66 instances per model, indicating a systemic characteristic of the summarisation task rather than a property of any single architecture. In each case, the model correctly identifies the domain and general nature of the rule violation but omits specific numeric thresholds or measured values (as defined in Section 3.3), reducing the precision of the output available to the human operator.

Lexical Deviation accounts for the remaining 39 failures (13.4%). In these instances, the reports covers the flagged risks but phrases them in weaker or more passive language than the reference, so the IMMEDIATE urgency marking is not carried. The underlying risks are addressed and the risk level correctly assessed, though the weakening phrasing could reduce the urgency conveyed to an operator relying on consistent flag prefixes. Granite 3.1 MoE records the highest Lexical Deviation count (13 events), meaning a greater proportion of its failures are vocabulary deviations rather than information omissions. Considered alongside its Compression Loss count of 60, Qwen3 has the lowest effective information loss rate among the four models.

Table 7
Final evaluation ($n = 100$): finalist model specifications and pipeline summary. All four models processed 100 supervisor invocations and 435 worker invocations (535 total LLM calls). Worker agent invocations reflect conditional domain activation across the 100 evaluated scenarios. The held-out evaluation set was sampled uniformly at random from the 32,065-row dataset, excluding all records used in Phase 1 screening.

Model	Architecture	Params (B)	Context (K)	Quantisation	Size (GB)
llama3.2:1b	Dense Transformer	1.0	128	fp16	2.5
granite3.1-moe:3b	MoE	3.0	128	q8_0	3.5
qwen3:4b	Hybrid-MoE	4.0	256	q8_0	4.3
granite4:tiny-h	Hybrid SSM-Transformer	7*	1000	–	4.2

* Granite 4 Tiny-H reports 7B nominal parameters; effective active parameters per forward pass are substantially lower due to sparse MoE and SSM routing. GPT-4o-mini is excluded from this stage as a deployment candidate; its screening scores are used as the cloud reference baseline throughout.

4.5. Fidelity to flags: comparison with a deterministic template baseline

The failure taxonomy measures divergence from the GPT-4o reference. As a complementary check that does not depend on a generative reference, a deterministic template baseline was constructed: each agent's report is produced by instantiating the fired rule reasons into fixed sentence templates, and the supervisor report is assembled by the same mechanism from the computed global score, risk band, and per-domain fired reasons, without any language model involvement at either level. Such reports are faithful to the flagged reasons by construction, carry no fabrication risk, and add negligible marginal cost;

Table 8
Final evaluation ($n = 100$ held-out scenarios): benchmark results for system reliability, reasoning quality, edge efficiency, and Risk-Weighted Scenario Fidelity (RWSF) for all four finalist models. Llama 3.2 1B is retained as the lower-bound reference representing the dense Transformer performance floor. All models were evaluated on the same 100 randomly sampled held-out scenarios using identical inference conditions via Ollama with strict JSON schema enforcement at temperature = 0.

Model	Reliability		Reasoning Quality		Edge Efficiency			Composite	
	JSON (%)	NC (%)	Smry.Sim	Rec.Sim	AGS (TPS)	VRAM (GB)	Time (s)	RWSF	EEI
llama3.2:1b[†]	87.3	100.0	0.671	0.887	73.1	5.16	390.7	0.287	0.104
granite3.1-moe:3b*	**100.0**	**100.0**	**0.856**	**0.889**	**112.2**	**5.43**[‡]	**380.4**	**0.289**	**0.109**
qwen3:4b	100.0	100.0	0.895	0.924	32.8	5.41	1225.9	0.302	0.085
granite4:tiny-h	100.0	100.0	0.878	0.894	48.6	5.34	1006.5	0.309	0.098

Column headers: JSON = Strict JSON schema adherence; NC = Negative constraint adherence (conformance to the prompt-level prohibition on conversational text); Smry.Sim = Agent summary cosine similarity (vs. GPT-4o-mini); Rec.Sim = Supervisor recommendation cosine similarity (vs. GPT-4o-mini); TPS = Average generation speed (tokens per second); VRAM = Peak memory usage (GB); Time = Total pipeline processing time (seconds); RWSF = Risk-Weighted Scenario Fidelity (a scenario-level composite metric penalising agent-domain divergences by operational risk weight, averaged across all 100 scenarios); EEI = Edge Efficiency Index.
* Granite 3.1 MoE identified as the preferred deployment candidate: highest throughput, highest EEI, lowest Compression Loss count, and lowest effective information-loss failure count after adjusting for non-critical Lexical Deviation events (see Section 4.7.).
[†] Llama 3.2 1B JSON adherence of 87.3% represents continued degradation from Phase 1 Batch 1 (91.6%) and Phase 1 Batch 2 (84.3%), confirming the sub-2B dense architecture's structural ceiling on format compliance at scale. Retained as the lower-bound reference only; not considered for deployment selection.
[‡] Granite 3.1 MoE records the highest VRAM consumption among the four finalist models (5.43 GB); all models operate within the 6 GB hard constraint and VRAM is a parity criterion rather than a discriminating factor in the deployment decision.

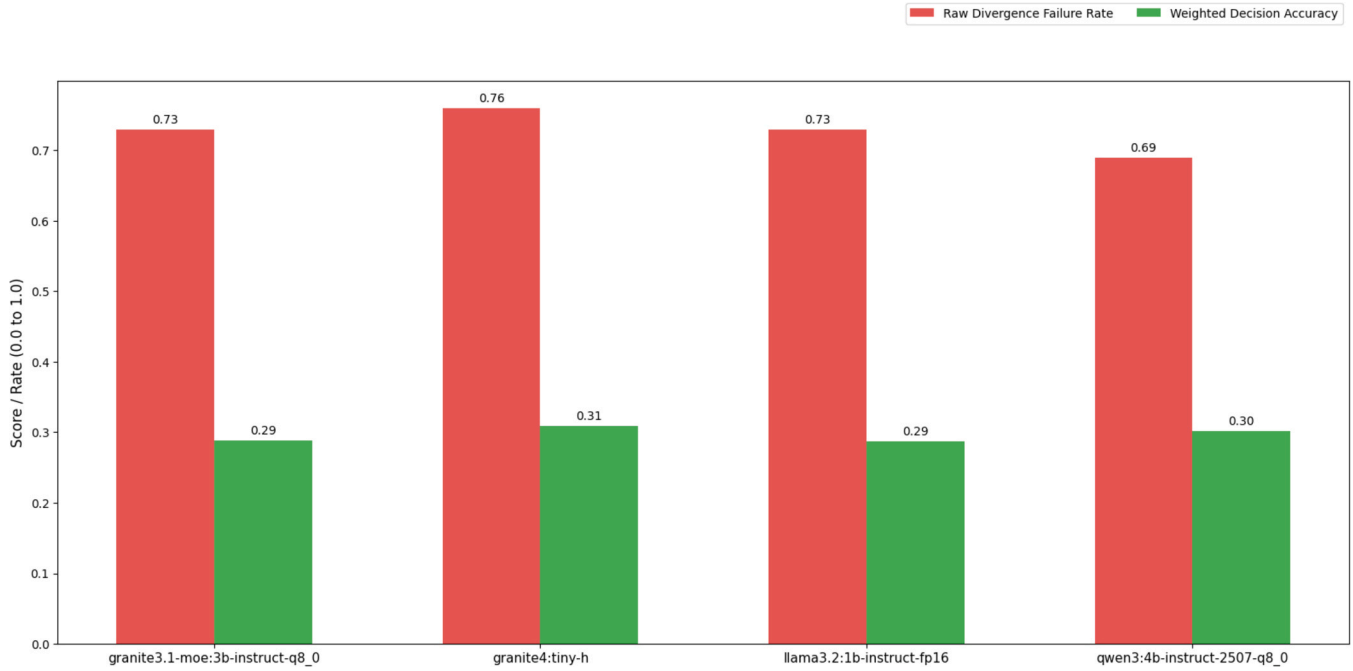


Fig. 5. Raw Divergence Failure Rate versus Risk-Weighted Scenario Fidelity (RWSF) across all four finalist models. Red bars represent the proportion of the 100 evaluated scenarios containing at least one agent-level divergence; green bars represent the corresponding RWSF score. The compression between red and green bars for each model reflects the proportion of failures absorbed by the risk-weighting correction.

what they cannot provide is fluent cross-domain synthesis or contextualisation.

Against this criterion, the deployed model's outputs were checked directly. An automated screen flagged candidate content beyond the fired reasons in 41 record-domain pairs covering 38 of the 100 evaluation records; all 41 were manually reviewed against the fired reasons and the underlying feature values. Two pairs were keyword artefacts in which no claim about the flagged feature was actually made, leaving content beyond the flags in 37 of 100 records. Classified by the strongest instance per record: 3 records contain a flat false claim, in which a comfortably safe value is asserted as a problem (supplier lead times of 4.1–7.2 days against a threshold of 10 described as suboptimal, and demand of 258–1314 units against a threshold of 5000 described as high); 17 records contain hedged consequences, in which a genuinely fired breach is described as posing potential knock-on effects on features whose values are safe (for example, severe weather posing risks across traffic, port, and customs operations); and 17 records are near-threshold, in which the mentioned value lies close to its bound (fuel consumption of 5.0–5.9 litres per hour against a lower bound of 5) or essentially at an exact equality sentinel (equipment availability of 0.002–0.012 against a rule that fires only at exactly zero).

This result can be interpreted in two complementary ways. The language layer adds what the fixed template does not attempt: cross-domain synthesis and operator-facing prioritisation, evidenced at the supervisor level. That synthesis carries a measurable and bounded cost, content beyond the flags in 37% of records, of which 3% rises to a flat false claim; under a strict fidelity-to-flags criterion the template baseline cannot introduce unflagged content and is superior on that axis alone. The near-threshold group includes three records in which the model's hedged caution was substantively supported by a value that the exact-

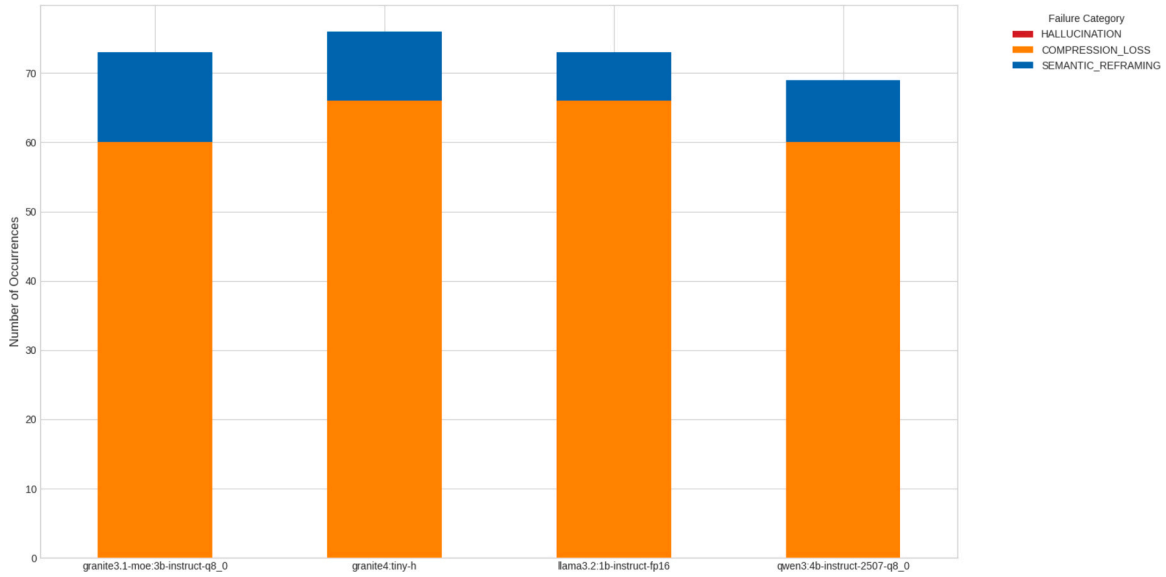


Fig. 6. Failure mode taxonomy across all four finalist models, classified by the automated LLM-as-a-judge procedure into three mutually exclusive root-cause categories: Compression Loss (orange), Lexical Deviation (blue), and Hallucination (red). No divergence was classified as fabrication across all 291 classified failures ($n = 100$ held-out evaluation).

Table 9
Failure mode taxonomy across all four finalist models, classified by the automated LLM-as-a-judge procedure into three mutually exclusive root-cause categories: Compression Loss, Lexical Deviation, and Hallucination. No divergence was classified as fabrication across all 291 classified failures ($n = 100$ held-out evaluation).

Model	Hallucination	Compression Loss	Lexical Deviation	Total
granite3.1-moe:3b	0	60	13	73
granite4:tiny-h	0	66	10	76
llama3.2:1b	0	66	7	73
qwen3:4b	0	60	9	69
Total	**0**	**252**	**39**	**291**

equality rule design could not flag. Two design implications follow. First, in every fabrication family, the features introduced are those recited in the agent's prompt but absent from the detected reasons, so constraining worker prompts to the detected reasons alone is the most direct mitigation. Second, adding tolerance bands to exact-equality sentinel rules would surface marginally clear values deterministically rather than leaving them to generative inference. These results concern the worker level of the deployed model.

At the supervisor level, template and generative reports were compared side by side on the 100-record Phase 1 screening sample (99 usable report pairs), using an automated screen over all pairs and a manual review of a 20-record subsample. The generative reports are the GPT-4o-mini reference reports for that sample, so the comparison also characterises reference generator behaviour; the recital padding described below therefore appears in both reference generators, GPT-4o-mini here and GPT-4o in the Phase 2 check (Section 5.5). The generative reports prioritise in every case (99 of 99, typically through IMMEDIATE-tagged recommendations), which the fixed template does not attempt, but they link findings across domains in 21 of 99 pairs, and the screen finds content beyond the fired reasons in 82 of 99. In the manual subsample, eight of twenty are clean generative wins, faithful to the flags and better prioritised, with no faithfulness cost; three show a weak or equal generative edge with hedged additions noted; two are contested; and seven lean towards the template on faithfulness grounds, where a domain that fired on a single breach was padded into a full failure narrative with invented values (a loading time of 2.9 h against a threshold of 4 described as excessive, and a supplier reliability of 0.996 described as anomalous). The padding follows the same prompt-recital mechanism observed at the worker level, most visibly in a recurring stock phrase enumerating the procurement prompt's full feature list, and it concentrates in domains that fired on a single breach, although exceptions in the subsample mean this is a tendency rather than a rule. The comparison isolates a contribution beyond linguistic presentation. The fixed template renders each domain independently and has no mechanism to relate a flag in one domain to a consequence in another, whereas the generative reports do so in 21 of the 99 pairs without the combinations being specified in advance. This supports the framework's position of confining generative synthesis to deterministically verified inputs, and it reinforces that per-record human review remains necessary before operational action (Section 5.5).

4.6. *Architectural analysis: KV-cache versus state-space memory compression*

Section 4.2. established that Granite 4 Tiny-H's Mamba-2 SSM layers apply selective compression of prior context rather than the lossless verbatim retention of KV-cache architectures, predicting a different failure distribution across the risk spectrum rather than uniformly better or worse numeric retention. The failure taxonomy data provides partial support for this prediction.

At the level of raw Compression Loss counts, Granite 4 records more failures (66) than Granite 3.1 MoE (60), contrary to any expectation that selective compression would reduce total information loss. The RWSF scores suggest a different pattern: Granite 4 achieves the highest RWSF in the evaluated set (0.309) compared to Granite 3.1 (0.289), and its failed rows retain more residual fidelity on average (0.091 per failed row versus 0.026). One interpretation consistent with the SSM architecture is that Granite 4's compression losses concentrate in lower-risk domains, leaving higher-risk domain outputs more faithfully preserved. By contrast, Granite 3.1's KV cache stores all tokens with equal lossless fidelity and applies no salience weighting across domains; its lower raw Compression Loss count is consistent with lossless retention of flagged content in the worker summaries that feed the fixed supervisor, since the content absent from a summary cannot surface in the recommendations. However, the RWSF difference of 0.02 sits within the narrow 0.29–0.31

range observed across all evaluated models. Without a per-domain failure breakdown confirming which specific domains are preserved, and without statistical significance testing at this sample size ($n = 100$), the selective-compression interpretation remains a hypothesis consistent with the data rather than a confirmed finding.

Throughput differs sharply between the two architectures. At 48.6 TPS versus 112.2 TPS, Granite 4 operates at less than half the generation speed of Granite 3.1, completing the 100-record Phase 2 pipeline in 1006 seconds compared to 380 s. Interpretation of these findings in the context of architecture selection is discussed in Section 5.

4.7. Deployment candidate selection

Across the three evaluation dimensions, Granite 3.1 MoE is selected as the preferred deployment candidate. Supporting metrics are summarised here; interpretation in the context of the research gaps is developed in Section 5.

At 112.2 TPS, Granite 3.1 MoE holds a 2.3x to 3.4x throughput advantage over Qwen3 4B (32.8 TPS) and Granite 4 Tiny-H (48.6 TPS), completing the 100-record pipeline in 380 s. On content retention, Granite 3.1 MoE and Qwen3 4B record identical Compression Loss counts (60 each), with Qwen3 marginally lower on Lexical Deviation (9 versus 13) producing a lower total failure count (69 versus 73). Qwen3 4B achieves higher semantic similarity (Agent Summary Similarity 0.895 versus 0.856; Supervisor Recommendation Similarity 0.924 versus 0.889). Granite 3.1 MoE's hardware footprint peaks at 5.43 GB within the 6 GB constraint and maintains 100% format reliability throughout. Granite 4 Tiny-H achieves the highest RWSF (0.309) but requires 1006 seconds to complete the same pipeline.

Synthesised via the Edge Efficiency Index (EEI), Granite 3.1 MoE achieves the highest overall operational score (0.109) compared to Granite 4 Tiny-H (0.098) and Qwen3 4B (0.085), confirming that its throughput advantage outweighs the marginal RWSF penalty and supporting its selection as the preferred deployment candidate.

4.8. Cost and latency analysis: hybrid versus fully cloud-based deployment

Phase 3 scaled the evaluation to $n = 1,000$ records, comparing a fully cloud-based configuration (GPT-4o-mini workers, GPT-3.5-Turbo supervisor) against the proposed hybrid edge-cloud configuration (Granite 3.1 MoE workers locally, GPT-3.5-Turbo supervisor). Both configurations processed 1000 supervisor invocations and 4326 worker invocations. Results are summarised in Tables 10 and 11.

The hybrid pipeline completed the workload in 3050.85 seconds, compared to 7134.57 seconds (approximately two hours) for the cloud-based configuration, a 2.3x reduction. Network round-trip latency across 5326 sequential API calls bottlenecked the cloud-based pipeline, which averaged 38.0 TPS. Offloading worker inference to Granite 3.1 MoE raised average generation speed to 110.7 TPS whilst keeping peak VRAM at 4.4 GB, well within the 6 GB constraint. Both configurations maintained 100% JSON and negative constraint adherence across all invocations.

Table 10
Workload and latency comparison at scale ($n = 1000$ scenarios). Both configurations successfully processed 5326 total LLM invocations (1000 supervisor calls and 4326 worker calls) with 100% strict schema adherence.

Configuration	Total Worker Tokens	Total Sup. Tokens	Peak VRAM (GB)	TPS	Processing Time (seconds)
cloud-based	596,130	424,305	–	38.0	7134.57
hybrid edge-cloud	881,308	406,453	4.4	110.7	3050.85

Note: Total tokens aggregate both input and output contexts. Cloud-based VRAM is omitted as inference occurs via cloud API. The hybrid configuration processes a higher worker token count (881,308 versus 596,130), reflecting differences in the Granite 3.1 MoE tokeniser relative to the OpenAI tokeniser; this incurs no additional cost since local inference carries no per-token charge.

Table 11
Operational cost comparison across the 1,000-scenario pipeline. Shifting worker agent inference to the local edge node yields a near 40% reduction in total API expenditure.

Configuration	Wkr. In ($)	Wkr. Out ($)	Sup. In ($)	Sup. Out ($)	Total ($)	Savings
cloud-based[a]	0.067	0.102	0.165	0.168	0.502	–
hybrid edge-cloud[b]	0.000	0.000	0.154	0.149	0.303	**39.6%**

[a] cloud-based: GPT-4o-mini (worker agents) + GPT-3.5-Turbo (supervisor).
[b] hybrid edge-cloud: Granite 3.1 MoE locally (worker agents) + GPT-3.5-Turbo (supervisor).
Note: Wkr. = worker agent; Sup. = supervisor. API costs reflect commercial pricing at the time of experiments. Local deployment of worker agents incurs zero per-call API cost.

At the per-record level, the hybrid pipeline averaged 3.05 s of end-to-end processing per record at $n = 1000$ (3.80 s at $n = 100$), corresponding to a sustained sequential throughput of approximately 1180 records per hour. Summed model-inference latency per record averaged 3.00 s (standard deviation 0.61, 95th percentile 3.95, maximum 9.29) at $n = 1000$ and 3.24 s (standard deviation 0.99, 95th percentile 4.30, maximum 8.76) at $n = 100$, so latency variation across records is modest and bounded. Because records are processed independently and no state is carried between them, delays do not accumulate across the pipeline; in a live stream, a backlog would form only if the sustained arrival rate exceeded this service rate. These figures are measured under sequential batch processing and are indicative of, rather than a substitute for, behaviour under live data arrival.

Accompanying the latency advantage is a reduction in API cost. As detailed in Table 11, offloading the six worker agents to the local edge node eliminates their token costs entirely. Total API expenditure drops from $0.502 to $0.303 per 1000 records, a saving of 39.6%, based on OpenAI API pricing as of 1 April 2026.

As record volume increases, the cost gap between the two configurations widens: the cloud-based configuration scales all six worker API calls proportionally with volume, whilst the hybrid configuration scales only the single supervisor call. For continuously operating monitoring systems processing growing record volumes, this structural difference in cost scaling strengthens the financial case for hybrid deployment over time.

This comparison is confined to API cost and latency under the stated pricing assumptions and should not be read as a total cost of ownership result. Local deployment carries cost dimensions the API comparison omits: the capital cost and depreciation of the edge hardware, its energy draws under continuous operation, and the maintenance, monitoring, and update overhead of a self-hosted model, set against the per-token price variability and data-egress considerations of the cloud path. Credible values for these components depend on deployment scale and site-specific energy and labour prices, so a full total-cost-of-ownership model is deferred to future work.

4.9. Sensitivity analysis

The rule thresholds, domain weights, and risk-band cut-offs are researcher defined (Table 2), so this subsection examines whether the reported behaviour depends on the specific values chosen. The analysis has two layers. The first re-evaluates the deterministic tier over the full dataset ($n = 32,065$ processed records) under nine perturbations: all rule thresholds shifted by ±10% and ±20% (features on bounded 0–1 scales and sentinel equality rules excluded), equal domain weights ($w_i = 1/6$), and classification cutoffs shifted by ±0.05 and ±0.10. The second

re-runs the full pipeline on the 100-record validation set for all four Phase 2 models under five perturbations plus baseline: thresholds ±20% (with the associated scoring ranges shifted alongside the rule thresholds), equal weights, and cut-offs ±0.10, using the same evaluation protocol as Phase 2.

Two patterns emerge at the deterministic tier (Table 12). First, record level flagging is saturated: 99.96% of records breach at least one of the twenty-one rules at baseline, and shifting every threshold by ±20% moves this by no more than 0.05 %age points. On this dataset, nearly every record triggers at least one rule somewhere across the six domains (a mean of 4.35 of the six agents fire per validation record), so the binary record-level flag is not a discriminating margin, and the per-domain invocation pattern is the operative quantity. Second, the risk-level distribution presented to the operator is materially sensitive to the banding parameters: shifting the cutoffs by ±0.05 reclassifies 11–14% of records, ±0.10 reclassifies 20–30%, and equal weights reclassify 10.5%. The Low, Moderate, and High shares are therefore primarily a function of the banding convention rather than of the rule thresholds, which is consistent with presenting the bands and weights as configurable conventions requiring recalibration per deployment rather than validated constants.

At the generative layer (Table 13), the baseline scenario reproduces the Phase 2 results to within 0.002, providing an internal consistency check on the re-run. Under ±20% threshold shifts, which change the set of fired rules and hence the content the models must summarise, agent summary similarity falls by at most 0.042 (Granite 4 at –20%) and supervisor recommendation similarity remains within 0.869–0.924 across all twenty-four runs. JSON adherence stays at 100% under every scenario for the three models of 3B parameters and above; only Llama 3.2 1B fluctuates (82.5–89.1%), consistent with its baseline reliability profile. Under weight and cut-off perturbations, supervisor similarity moves by less than 0.005. The ranking of the four models on reasoning quality and reliability is identical in every scenario, so the deployment-candidate selection does not rest on the specific parameter values chosen. The analysis varies one parameter family at a time around the deployed configuration; joint perturbations and recalibration to other datasets are not explored, and the Layer 1 findings describe this dataset's distributions rather than a general property of the rule set.

Table 12
Layer 1 sensitivity: deterministic tier re-evaluated over the full dataset (n = 32,065) under each perturbation. Stability is the share of records retaining their baseline risk class. Threshold shifts alter rule-violation flagging only; risk scores are computed from separately parameterised scoring ranges, so mean S_{global} is unchanged by construction under those scenarios.

Perturbation	Flag rate (%)	Mean S_{global}	Low / Mod / High (%)	Stability (%)
Baseline	99.96	0.7423	0.1 / 35.0 / 64.9	100.0
Thresholds + 10%	99.97	0.7423	0.1 / 35.0 / 64.9	100.0
Thresholds + 20%	99.95	0.7423	0.1 / 35.0 / 64.9	100.0
Thresholds −10%	99.99	0.7423	0.1 / 35.0 / 64.9	100.0
Thresholds −20%	100.0	0.7423	0.1 / 35.0 / 64.9	100.0
Equal weights	99.96	0.7529	0.0 / 31.9 / 68.1	89.47
Cut-offs + 0.05	99.96	0.7423	0.3 / 49.0 / 50.7	85.59
Cut-offs + 0.10	99.96	0.7423	0.9 / 63.3 / 35.8	70.10
Cut-offs −0.05	99.96	0.7423	0.0 / 23.8 / 76.2	88.58
Cut-offs −0.10	99.96	0.7423	0.0 / 15.1 / 84.9	79.88

Table 13
Layer 2 sensitivity: full pipeline re-run on the 100-record validation set under each perturbation. Smry.Sim = agent summary cosine similarity; Rec.Sim = supervisor recommendation cosine similarity; JSON = strict schema adherence. Under weight and cut-off perturbations the fired rules are identical to baseline, so worker summaries are unchanged and only the supervisor's risk framing shifts.

Scenario	Smry.Sim	Rec.Sim	JSON (%)
llama3.2:1b	0.671	0.887	87.3
Baseline			
Thresholds + 20%	0.656	0.878	89.1
Thresholds –20%	0.644	0.869	82.5
Equal weights	0.671	0.888	87.3
Cut-offs + 0.10	0.671	0.888	87.3
Cut-offs –0.10	0.671	0.886	87.3
granite3.1-moe:3b	0.856	0.888	100.0
Baseline			
Thresholds + 20%	0.828	0.872	100.0
Thresholds –20%	0.832	0.875	100.0
Equal weights	0.856	0.893	100.0
Cut-offs + 0.10	0.856	0.893	100.0
Cut-offs –0.10	0.856	0.889	100.0
qwen3:4b	0.895	0.923	100.0
Baseline			
Thresholds + 20%	0.864	0.900	100.0
Thresholds –20%	0.856	0.910	100.0
Equal weights	0.896	0.924	100.0
Cut-offs + 0.10	0.895	0.923	100.0
Cut-offs –0.10	0.895	0.921	100.0
granite4:tiny-h	0.877	0.893	100.0
Baseline			
Thresholds + 20%	0.842	0.874	100.0
Thresholds –20%	0.835	0.885	100.0
Equal weights	0.878	0.894	100.0
Cut-offs + 0.10	0.874	0.892	100.0
Cut-offs –0.10	0.875	0.894	100.0

5. Discussion

5.1. Addressing the research gaps

On the operational monitoring latency gap, the results demonstrate that deterministic rule-violation detection and generative natural language reasoning can operate within latency budgets consistent with continuous monitoring when embedded in a hierarchical multi-agent system. Granite 3.1 MoE completes the 100-record pipeline in 380 s and the 1,000-record pipeline in 3050 seconds, establishing a throughput baseline compatible with continuous monitoring workloads, though the present evaluation is batch-mode and does not constitute a live real-time deployment test. At the architectural level, Tier 1 operates without language model involvement; Tier 2 is invoked conditionally only on confirmed rule violations; and Tier 3, proposed as a design component for future empirical validation, delivers the complete traceable report to the human operator. Because each record is processed independently, no upstream latency accumulates across the pipeline. This addresses the monitoring stage of SCRM that Ordibazar et al. [14] identify as the least developed in the current AI-based SCRM literature, extending beyond the strategic and planning orientation of existing LLM-SCRM frameworks [7,19,37].

On the hybrid edge-cloud deployment gap, the benchmarking results provide empirical evidence that local SLMs in the 1–7 billion parameter range are viable components for agentic supply chain systems. Because detection is deterministic, every evaluated model receives an identical set of rule violation flags from the Tier 1 rule engine, so flag reproduction is a property of the rule configuration rather than a model level result; the SLMs' contribution lies in the generative synthesis built on that fixed input, not in detection accuracy. By converting raw sensor readings into structured, factual findings before any language model is invoked, the architecture removes detection reliability as a variable dependent on model size, extending the argument of Belcak et al. [20]

that SLMs are suitable for specialised, repetitive agentic tasks to the specific context of supply chain risk monitoring. At $n = 1,000$ records, the hybrid edge-cloud configuration reduces API cost by 39.6% relative to the fully cloud-based deployment, with the cost gap widening as record volume increases.

On the transparency and decision-support gap, the rule-first detection design ensures that every risk flag and score at the detection layer is the product of explicit, inspectable rules, making each alert auditable by the operator. Natural language summaries generated in Tier 2 express confirmed findings in terms that practitioners can read, evaluate, and challenge. This responds to the recommendations of Ordibazar et al. [14] for explainability methods and practitioner-facing dashboards in AI-based SCRM, and aligns with the transparency principles of Vössing et al. [11], who find that making AI reasoning visible to operators increases trust and collaborative task outcomes. Rather than asking the language model to disclose its own uncertainty, which Vössing et al. [11] identify as reducing perceived system reliability, uncertainty is absorbed by the deterministic layer and the generative layer presents only confirmed findings.

Model selection for hybrid edge-cloud deployment proves to be a multi-criteria decision rather than a single-metric optimisation. Granite 4 Tiny-H achieves the highest RWSF (0.309), and the pattern of its failed rows retaining more residual fidelity than those of the KV-cache models is consistent with the prediction that SSM-based compression would selectively preserve higher-salience content, though the RWSF difference (0.02) falls within the narrow band observed across all models. The KV-cache architectures (Granite 3.1 MoE and Qwen3 4B) apply no such weighting and rely on lossless storage, retaining the full input for carry-over into the summaries that feed the supervisor. In this evaluation, neither the RWSF advantage of Granite 4 nor the marginal reasoning-quality advantage of Qwen3 offsets their respective throughput costs (48.6 TPS and 32.8 TPS versus 112.2 TPS for Granite 3.1). Granite 3.1 MoE is preferred because it maintains equal content retention to Qwen3, a throughput advantage of 2.3x to 3.4x over the other candidates, 100% format reliability, and a memory footprint within the 6 GB VRAM constraint of edge-representative hardware. Residual reasoning-quality gaps could be narrowed through domain-specific fine-tuning [39] without sacrificing the throughput that makes the model operationally viable. More broadly, architecture selection between hybrid SSM-Transformer and standard MoE depends on the memory contract and latency tolerance of the target deployment environment, not on any single performance metric.

5.2. Theoretical contributions

The first contribution is the formalisation of rule-first detection design as the mechanism enabling low-latency operational monitoring in agentic supply chain systems. Prior work in LLM-based SCRM has predominantly positioned language models as the primary reasoning layer, with structured data or rules as supplementary inputs: Zhao et al. [19] use LLMs as the primary risk identification engine operating on unstructured text; Quan et al. [37] use LLMs as the primary interpreter of risk narratives within the MARS multi-agent framework; Zheng et al. [38] integrate LLMs as the primary reasoning component within a decentralised agent-based system. The proposed framework reassigns these roles: deterministic rules are the primary detection mechanism, and language models serve as a secondary explanation layer. Detection requires precision, speed, and verifiability; explanation requires contextual reasoning and language fluency. Assigning each capability to the layer best suited to it produces two operational benefits at once: the deterministic path delivers latency low enough for continuous monitoring, and the generative path delivers practitioner-facing explanations without being placed on the critical detection path. To the best of our knowledge, this separation is not made explicitly in prior frameworks.

A second contribution is the empirical demonstration of task-matched model selection producing a differentiated performance profile across evaluation dimensions at the system level. No single model dominates all three dimensions: Granite 3.1 MoE leads on throughput (112.2 TPS), completing the 100-record pipeline in 380 s; Qwen3 4B leads on reasoning quality, recording the highest Supervisor Recommendation Similarity (0.924) of any local candidate; and Granite 4 Tiny-H achieves the highest RWSF (0.309), though RWSF scores across all four models compress into a narrow band (0.287–0.309) and the metric did not produce strong discriminating power in this experimental setup. The stronger evidence for task-matched selection comes from the throughput and failure-type differentiation: architectural choices that favour speed (MoE routing in Granite 3.1) versus reasoning depth (dense attention in Qwen3) versus memory efficiency (SSM compression in Granite 4) produce measurably different operational trade-offs. This supports the argument for heterogeneous multi-agent systems in which model selection is task-driven rather than scale-driven, extending the position of Belcak et al. [20] with empirical evidence from a supply chain risk monitoring context.

A third contribution is the operationalisation of transparency by design as a mechanism for building human-AI trust in operational decision support. Whilst the transparency literature has largely focused on post-hoc explainability methods applied to trained machine learning classifiers [11], this framework achieves transparency through ex-ante design: rules are defined before deployment, are human-readable, and are the exclusive source of rule violation classifications. The distinction has direct bearing on AI governance in supply chain operations, where the ability to audit the basis of an automated risk flag is a prerequisite for practitioner trust and action [10]. Hendriksen [33] further argues that AI integration in SCM is not only a technical but also a social process shaped by human sensemaking, aided when the system's reasoning is visible and inspectable rather than opaque. The failure-mode analysis provides partial grounding for this claim at the detection layer, where rule-based classification excludes hallucination by construction. It does not extend to the generative layer: the fabrication measure compares outputs to the GPT-4o reference rather than to the flagged reasons, and a direct check against those reasons finds content beyond the flags in 37 of 100 records for the deployed model and in 86 of 100 for the reference itself (Section 4.5). In live operation a generative output can still omit a detail, misstate a risk, or over-extend a recommendation, particularly under noisy inputs or co-occurring rule violations, so human review of generated output remains necessary before any operational action, with the Tier 3 operator serving as that verification point by design.

5.3. Practical implications for operational supply chain managers

Operational supply chain managers responsible for day-to-day logistics execution, monitoring active shipments, responding to supplier failures, and managing workforce-related risk signals, are the primary audience for this framework. Kolbjørnsrud et al. [10] identify this population as exhibiting the greatest reluctance to act on AI-generated recommendations, not because they doubt AI's potential, but because they cannot interrogate or verify the basis of what is recommended to them.

The framework addresses this directly. Every risk flag and score is anchored to an explicit, inspectable deterministic rule. Tier 2 natural language summaries express only confirmed deterministic findings. Together these provide a traceable chain of evidence from sensor reading to rule trigger, to natural language summary, to supervisor report. Practitioners can therefore scrutinise, accept, or override any recommendation with confidence. The human operator occupies the final and authoritative tier of the proposed workflow: the system maintains a monitoring frequency and breadth that manual review cannot match, whilst the practitioner retains the contextual authority and operational accountability that no automated system can carry. This division of labour is consistent with the human-AI collaboration model proposed by Boone et al. [22], in which AI enhances rather than replaces human expertise.

For organisations evaluating deployment, the cost analysis provides a practical reference point. The 39.6% API cost reduction at $n = 1,000$ records translate to a widening financial advantage at operational scale as record volume grows. Where data privacy constraints or intermittent connectivity apply, the worker agent tier can run entirely on local hardware using open-source SLMs, with no data transmission to third-party services, addressing a barrier to LLM adoption in supply chain operations.

The risk banding that determines what an operator sees is a management lever rather than a technical constant. Shifting the band cut-offs by ±0.10 reclassifies 20–30% of records whilst leaving detection itself untouched (Section 4.9), so band placement should be set deliberately against an organisation's alert-handling capacity and risk appetite and revisited as monitoring volumes change. A second consideration concerns what the generative tier can be expected to deliver. Disruption propagation across functions, the ripple effect analysed by Ivanov et al. [1], is precisely what a supervisor tier is positioned to surface, yet generative reports linked findings across domains in only 21 of 99 report pairs in the baseline comparison (Section 4.5), and the evaluation dataset's lack of cross-feature correlation means this capability could not be meaningfully exercised or tested here. Recalibration guidance for new operational contexts and evidence from live deployment likewise remain future work (Section 5.5).

5.4. Illustrative interface design: the veriflow dashboard

VeriFlow is a dashboard prototype, populated with synthetic illustrative data, that demonstrates how the framework's outputs could be presented to a practitioner in a deployed setting. It is not a component of the empirical evaluation.

Fig. 7 presents the overview panel, showing how supervisor-level outputs can be aggregated into operational KPIs covering analysed shipments, a breakdown of rule violations by agent domain, and temporal risk trends displaying the monthly distribution of Low, Moderate, and High classifications. Fig. 8 presents the shipment detail and geospatial tracking panels. At the record level, the computed risk score, rule violation source, and supervisor narrative are surfaced for operator review, whilst a geospatial risk distribution map supports rapid identification of spatial clustering among high-risk records.

5.5. Limitations

The following points outline the limitations of this study. Empirical validation is based on a single dataset from a Southern California logistics network [41], and the deterministic rule thresholds were established drawing on domain knowledge and the distributional properties of this specific dataset. Whilst the thresholds and domain decomposition are configurable, no recalibration methodology is provided for different operational environments. Whether the current threshold values and domain boundaries remain appropriate for materially different contexts, such as cold-chain pharmaceutical distribution, international maritime freight, or manufacturing just-in-time networks, requires validation on independent data.

The retained features show no meaningful linear correlation (all $|r| < 0.015$; Section 3.1), a structure that does not reflect the cross-domain dependencies typical of live operational data and that bounds the external validity of the evaluation. The framework does not rely on feature independence as a modelling assumption: each agent monitors its assigned domain against fixed thresholds regardless of conditions elsewhere, and correlated rule violations appearing simultaneously across domains would naturally elevate multiple agent scores, producing a higher S_{global} and a correspondingly escalated supervisor response.



Fig. 7. VeriFlow dashboard prototype (Overview panel). Supervisor-level outputs are aggregated into operational KPIs, temporal risk trends, and domain-specific rule violation distributions for operator review.

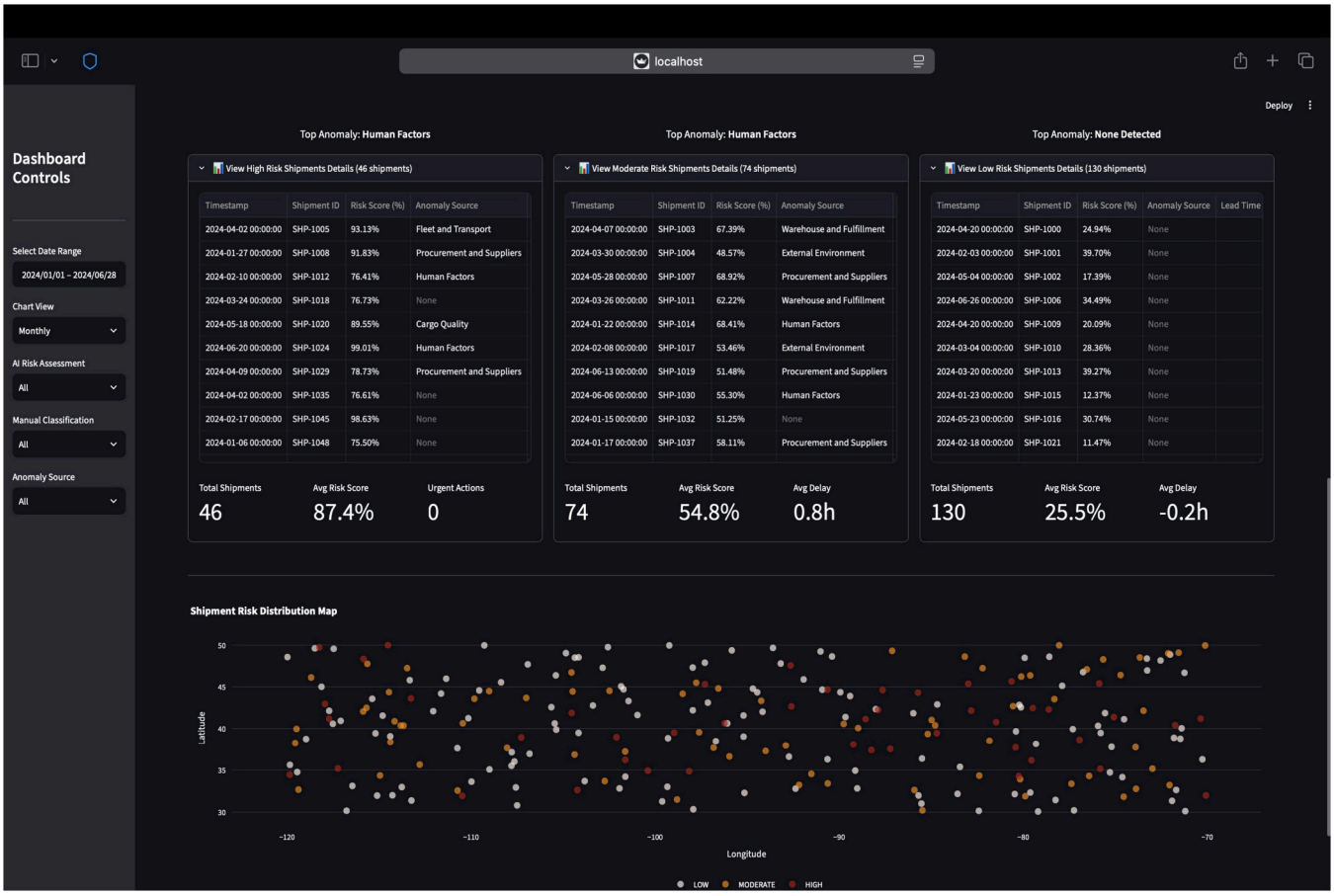


Fig. 8. VeriFlow dashboard prototype (Shipment detail and geospatial panels). Record-level multi-agent outputs, including risk scores and natural language rationales, are surfaced alongside geospatial tracking to support human-in-the-loop review.

Validation against datasets with realistic inter-domain dependencies would test whether this property holds under more representative conditions.

The ground truth construction and failure classification processes described in Sections 3.3.1 and 3.3 both rely on LLM-generated outputs reviewed by two members of the research team without formal inter-rater reliability. A direct check of the reference outputs against the deterministically flagged reasons shows that the generative references introduce content not present in the flags, which the failure taxonomy would classify as fabrication, in 86 of the 100 evaluation records; all 128 flagged record-domain pairs were manually reviewed against the fired reasons and every one is a genuine mention of unflagged features, so the figure is validated to the same standard as the deployed model's. By a screen of the same design, the deployed model's outputs contain unflagged content in 38 of 100 records (37 on manual validation; Section 4.5), so the reference generator introduces content beyond the flags at a higher rate than the model evaluated against it. Two consequences follow for the evaluation. The failure classifier scores each model output by its divergence from these references, so it measures agreement with the reference rather than fidelity to the flags, and it cannot detect fabrication that a model shares with the reference. A contributing mechanism is visible in the manual check of the deployed model's outputs (Section 4.5): the worker prompts name each agent's monitored data categories before supplying the detected reasons, and in every fabrication family identified, the features introduced are those recited in the prompt but absent from the flags. The same recital pattern appears in the reference outputs, most visibly in recurring stock sentences enumerating an agent's full feature list; constraining the prompt to the detected reasons is a direction for further work. The reported absence of fabrication should therefore be read as the absence of divergence from-reference fabrication under the tested conditions, not as evidence that outputs remained within the flagged facts. In a fully deployed system processing continuous data at scale, per-output human review would not be feasible; validating the framework's outputs under those conditions, through practitioner feedback in a live operational setting, including conditions with noisy inputs and multiple co-occurring risks, remains a priority for future work.

Cosine similarity against ground truth narratives, whilst useful for ranking models, does not capture all dimensions of practical utility. Whether the natural language outputs improve operator decision quality, reduce cognitive load, or increase trust relative to structured tabular outputs is an empirical question requiring a formal user study with supply chain practitioners.

Risk-Weighted Scenario Fidelity (RWSF) weighting failures by operational importance, captures a property that raw failure counts miss. It did not, however, produce strong discriminating power in this experimental setup. All four models scored within a range of 0.022 (0.287–0.309), narrower than the spread in raw failure rates. The compression-hits-broadly pattern observed under exact-match divergence analysis, in which most activated domains in a failed row diverge simultaneously, causes the metric to compress into a band driven primarily by each model's clean-row rate. The deployment decision was ultimately driven by throughput, format reliability, and failure-type analysis rather than by RWSF. Whether RWSF would discriminate more effectively under a partial-match or graded divergence criterion, or on datasets where compression events affect individual domains rather than entire rows, remains an open question for future evaluation design.

As a proof-of-concept, the framework carries limitations that would need to be addressed before production deployment. Static rule configurations would require active maintenance as operational baselines evolve, and the overhead of this maintenance is not addressed in the current study. The VeriFlow interface is a design prototype only and has not been evaluated for usability, comprehension, or operator acceptance.

A further design-level limitation arises from the relationship between continuous domain scores and binary rule-violation flags. A domain exhibiting significant stress reflected in a high continuous score may not trigger the anomaly flag if no individual variable crosses its specific threshold, resulting in no natural language explanation being generated for that domain despite its contribution to the global risk score. Whilst this is consistent with the intended rule-based design, it creates a partial transparency gap for sub-threshold domain stress. A secondary warning mechanism that generates a cautionary note when a domain score exceeds a defined warning threshold without triggering the full rule violation flag would close this gap.

Tier 3, the human decision layer, was not empirically evaluated. Its design is grounded in human-AI collaboration literature [10,11], but operational effectiveness, including operator response time, decision quality, and override behaviour, remains to be validated in a live deployment setting with supply chain practitioners.

6. Conclusion

This paper presented the design of a batch-mode proof-of-concept evaluation of a hierarchical multi-agent framework for proactive supply chain risk monitoring, integrating deterministic rule violation detection with hybrid edge-cloud SLM and LLM deployment to produce structured, traceable risk reports for human operator review, a design consistent with the human-centric orientation of Supply Chain 5.0. The framework addresses three gaps identified in the supply chain AI literature: the operational monitoring latency gap, the hybrid edge-cloud deployment gap, and the transparency and decision-support gap. Three core findings emerge from the empirical evaluation conducted on samples drawn from a 32,065-record logistics dataset. Because detection is deterministic, every model reproduces the same flags by construction, so flag detection is a property of the rule configuration rather than a model-level result. SLM-based generative synthesis produces contextually coherent risk summaries; among outputs diverging from the GPT-4o reference, none were classified as fabrication, though this measure compares outputs to that reference rather than to the flagged reasons (Section 5.5); Compression Loss, the omission of critical items carried by the reference report, accounts for 86.6% of all classified divergences. A direct check against the flagged reasons finds content beyond the flags in 37% of records, of which 3% contain a flat false claim (Section 4.5). Granite 3.1 MoE emerges as the preferred deployment candidate, combining the highest throughput (112.2 TPS), 100% format reliability, and the lowest effective information loss rate among the evaluated models. At scale ($n = 1,000$), the hybrid edge-cloud configuration reduces API cost by 39.6% and processing time by 57% relative to the fully cloud-based alternative, with the cost gap widening as record volume increases. These findings support three theoretical contributions formalised in the discussion: rule-first detection design, which positions deterministic rules as the primary detection mechanism and language models as a secondary explanation layer; task-matched model selection, which supports selection driven by task characteristics rather than parameter scale; and transparency by design, which achieves human-AI trust through ex-ante rule design rather than post-hoc explainability. The proof-of-concept scope means that generalisation to other logistics contexts requires further validation; subject to this, the three principles offer a replicable design pattern for AI-enabled operational monitoring systems where detection speed, output traceability, and practitioner trust are concurrent requirements.

Future research will focus on deployment against live data streams from operational logistics systems as the most immediate next step, as it would validate detection latency under genuine real-time conditions and surface failure modes that batch evaluation cannot expose. Portability of the deterministic rule schemas and SLM prompt configurations across different operational contexts remains untested. Prospective validation on an independent dataset from a materially different logistics environment, such as cold-chain pharmaceutical distribution, international maritime freight, or manufacturing just-in-time networks, would strengthen the basis for broader deployment. A formal user study involving operational supply chain managers, measuring decision confidence, cognitive load, and decision quality relative to baseline support tools, would provide direct empirical grounding for the transparency and trust claims made in this paper and would evaluate the VeriFlow interface for usability and operator acceptance. Static rule thresholds would require active maintenance in a production environment as operational baselines evolve. Incorporating mechanisms that adapt thresholds automatically in response to distributional shift would reduce this maintenance overhead and improve long-term operational responsiveness. Compression Loss, the dominant failure mode, points to a need for evaluation metrics that go beyond cosine similarity. Developing instruments that assess whether generated summaries faithfully represent the triggered deterministic rules, without omitting critical rule violations or introducing unsupported inferences, would provide a more precise measure of generative fidelity for this class of task. Risk-Weighted Scenario Fidelity and the Edge Efficiency Index are proposed here as exploratory indicators for this implementation and establishing whether either holds value as a general measure would require validation against practitioner judgement and replication across independent deployments. The local execution model is architecturally compatible with IoT-integrated environments where continuous sensor streams, data privacy requirements, and intermittent connectivity make cloud-dependent inference impractical. Extension to broader industrial contexts such as production line monitoring and digital twin integration [4] would test the generalisability of the architecture, subject to domain-specific validation.

CRediT authorship contribution statement

Natalia Hartono: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kevin Nathanael:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Voicu Ion Sucala:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Martino Luis:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Elioenai Sitepu:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Natalia Hartono reports article publishing charges was provided by University of Exeter. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This research received no external funding. For the purpose of open access, the author has applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising from this submission.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.sca.2026.100239.

Data availability

This study uses a publicly available dataset; the source is cited in the manuscript. The code is available on https://github.com/NataliaHartonoFung/supply-chain-risk-and-decision.git.

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